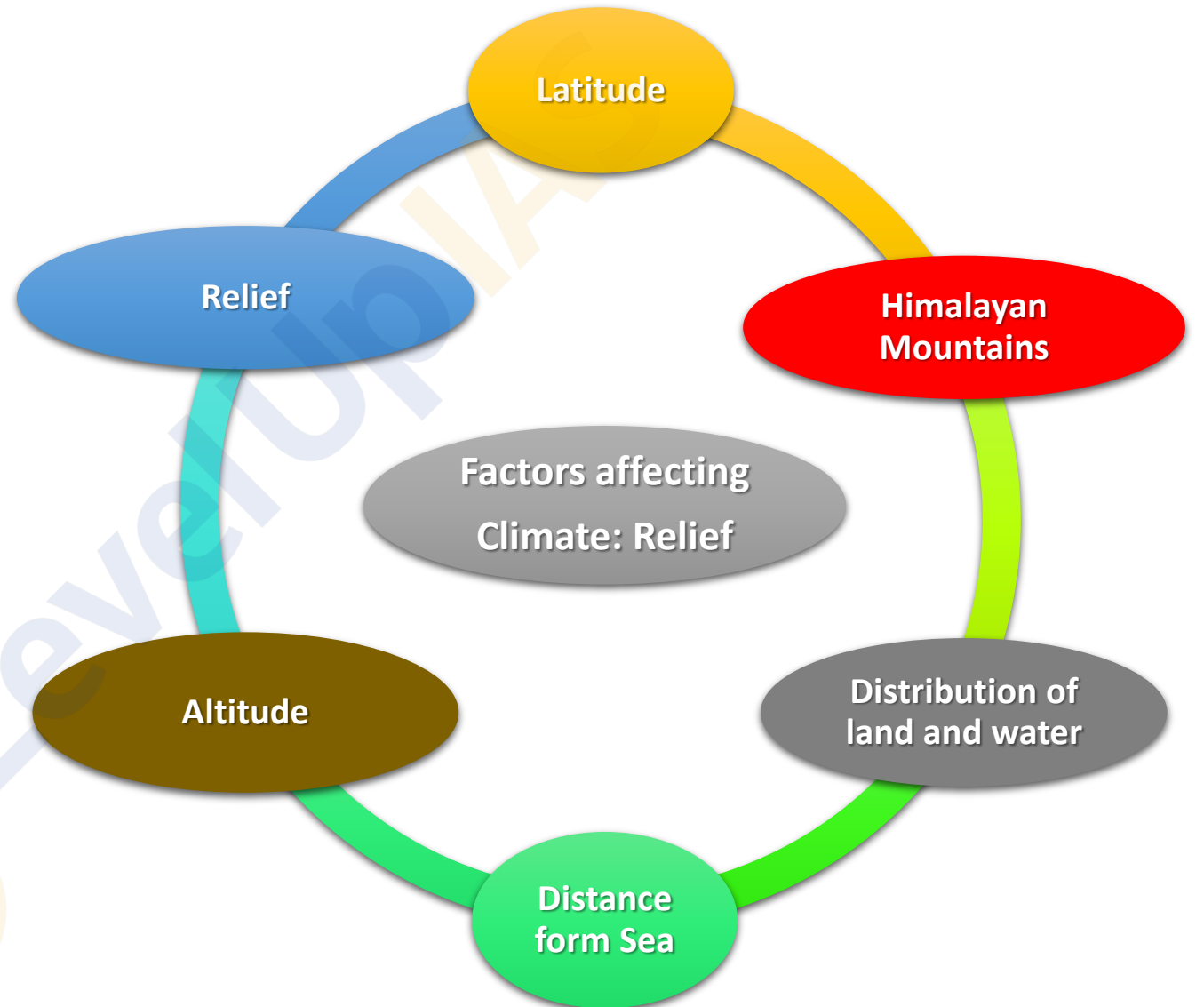
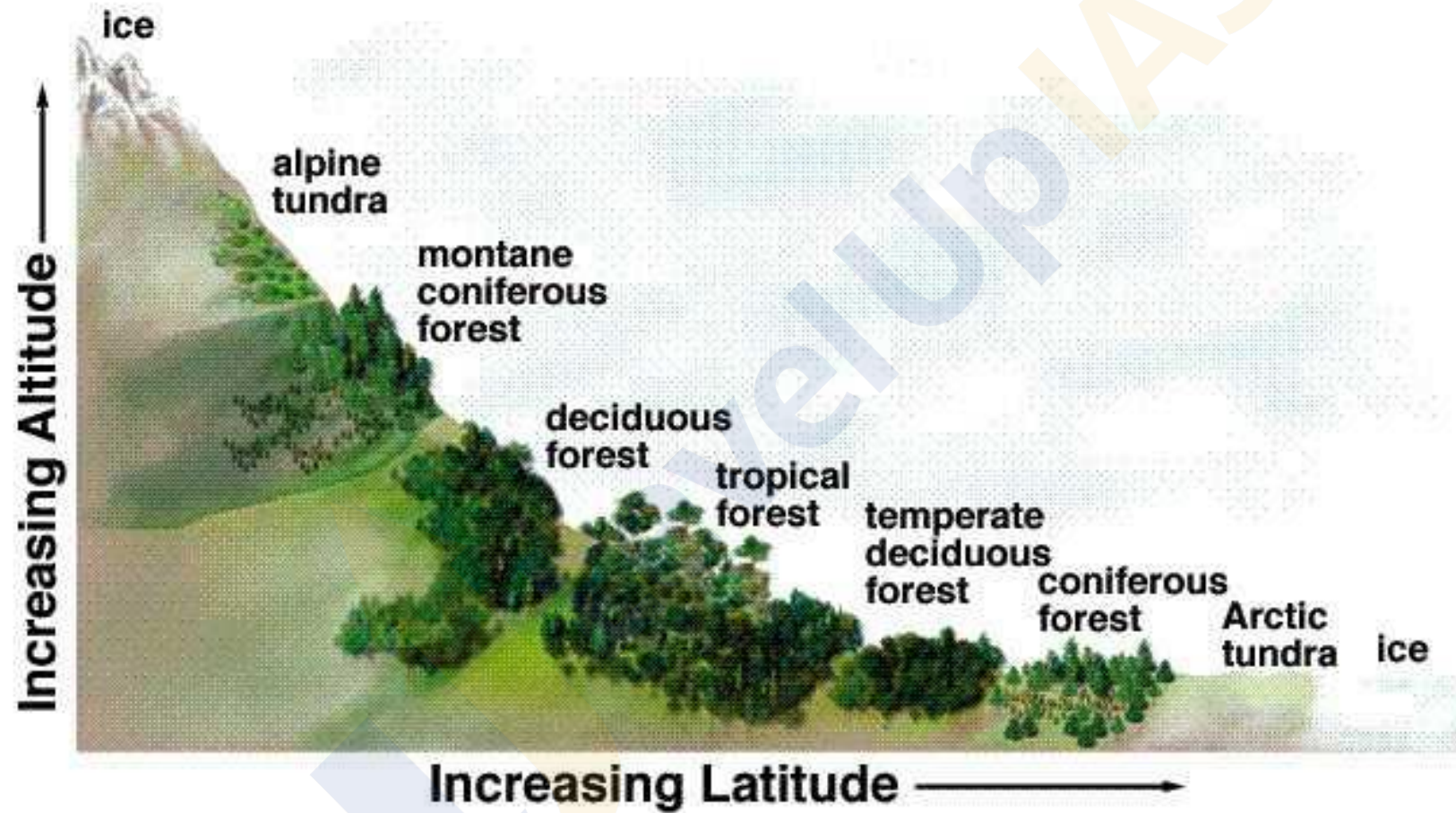


Geography: Climate and Seasons: India

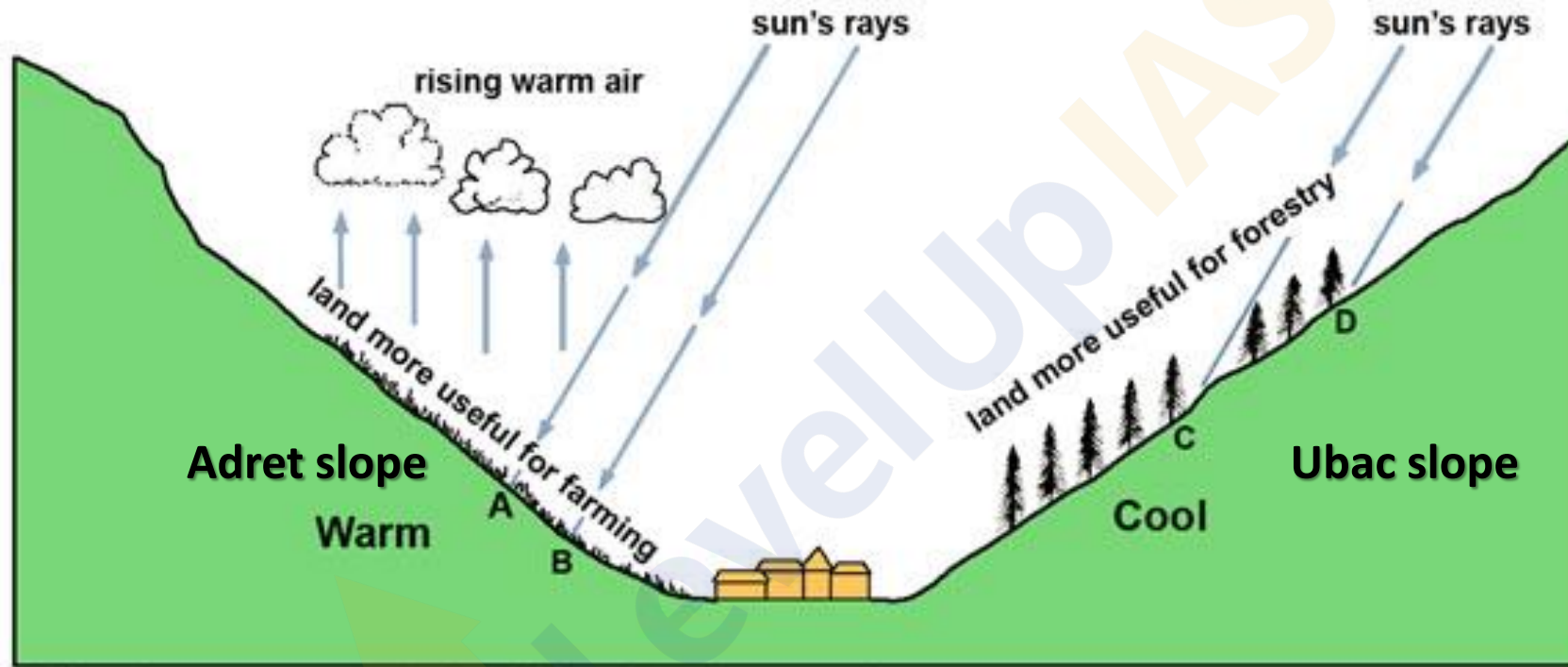
- DIMPLE NANKANI



Altitudinal and Latitudinal variation

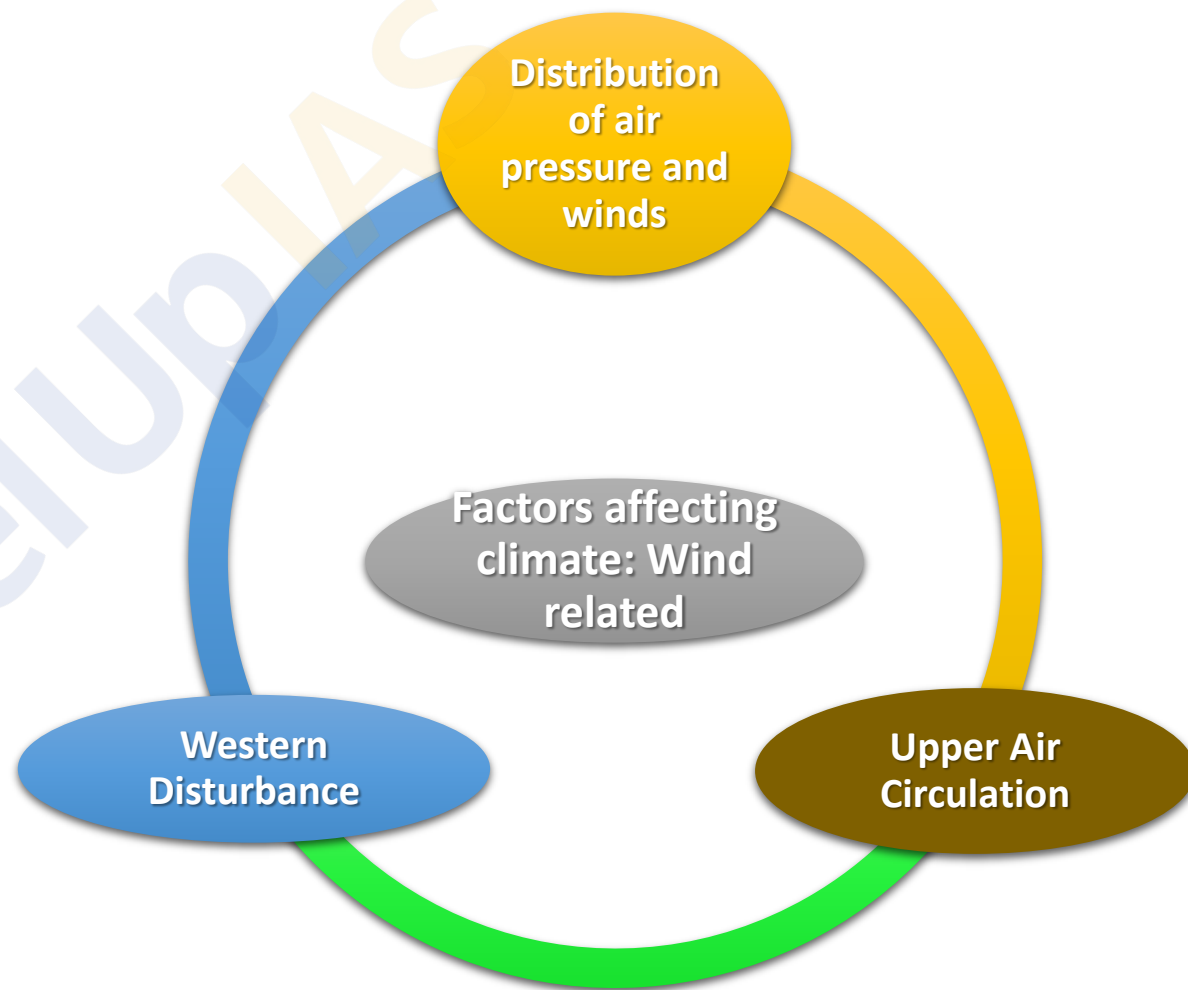


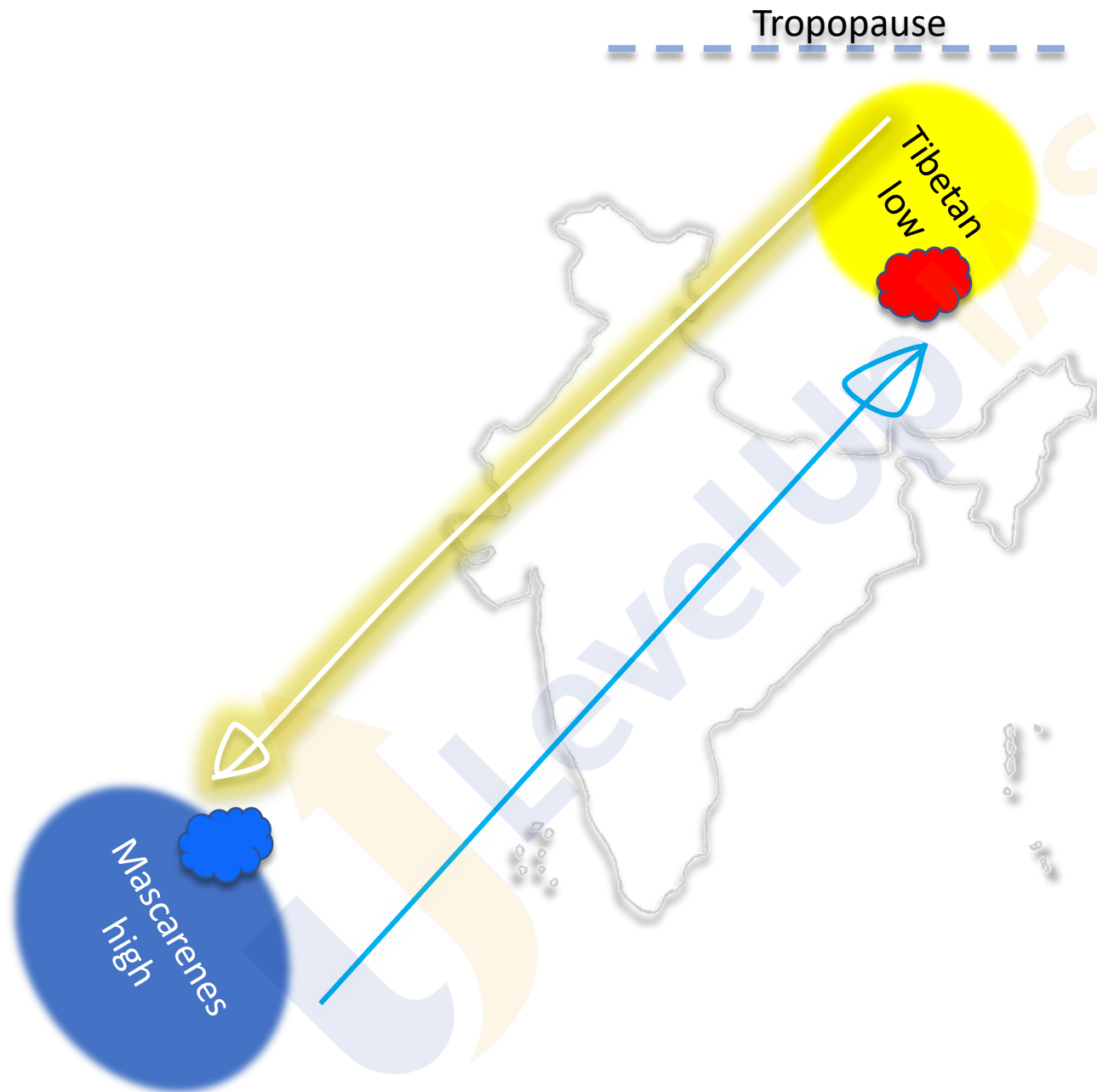
The uneven heating in a valley

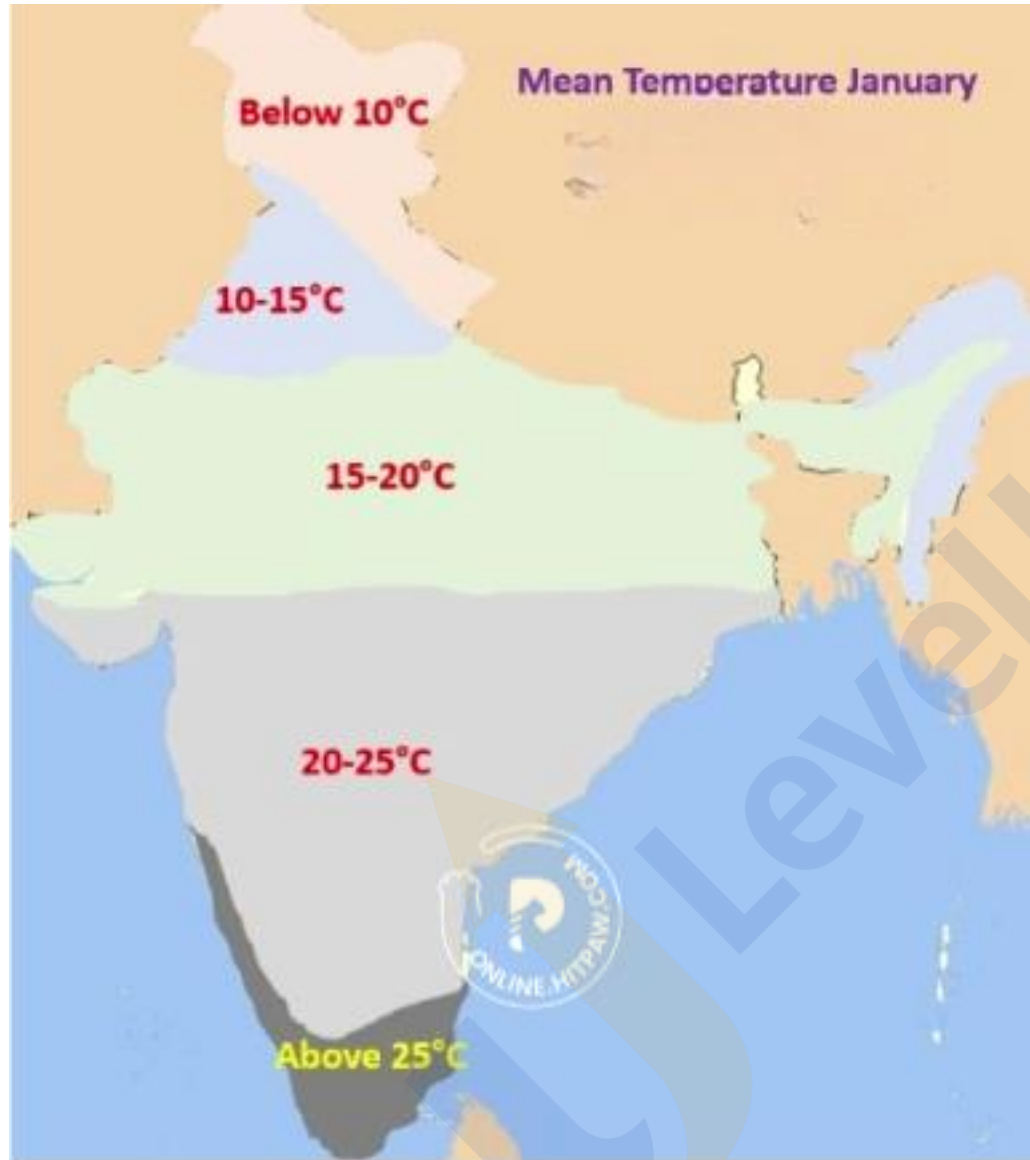


A-B sun's rays concentrated at high angle to the ground
C-D sun's rays concentrated at low angle to the ground

Slope Condition

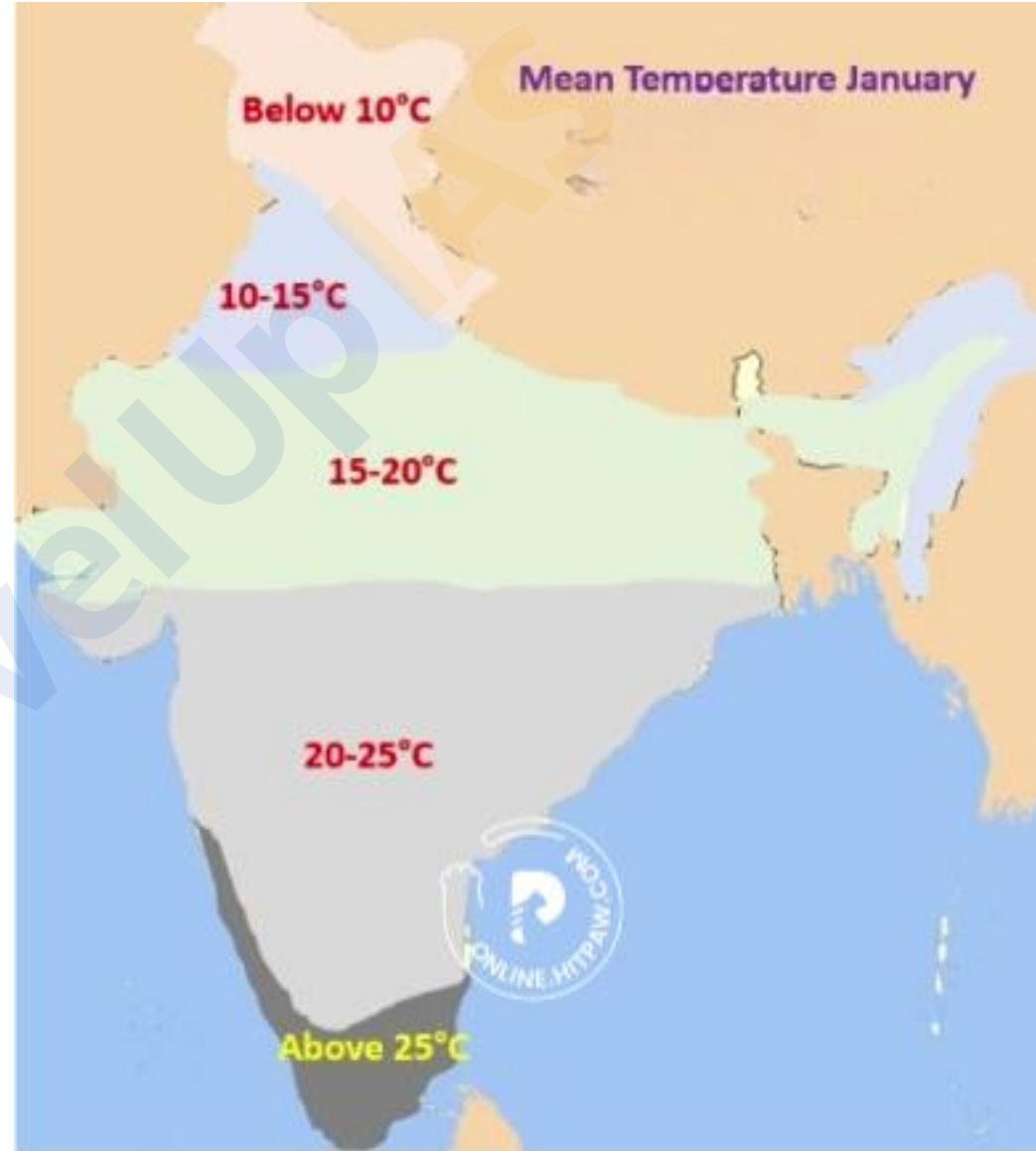






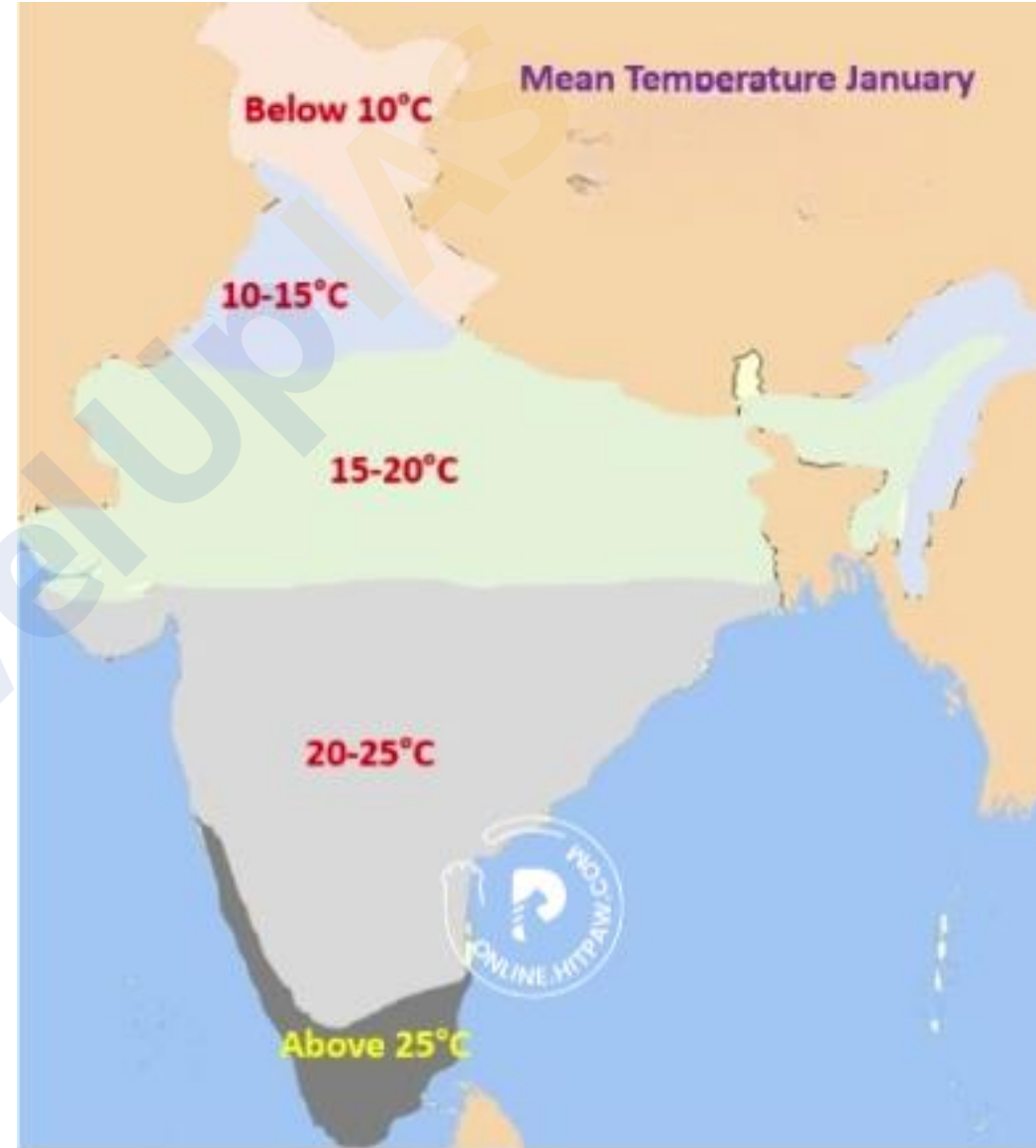
Climate of India: Winter Season (Nov-Jan)

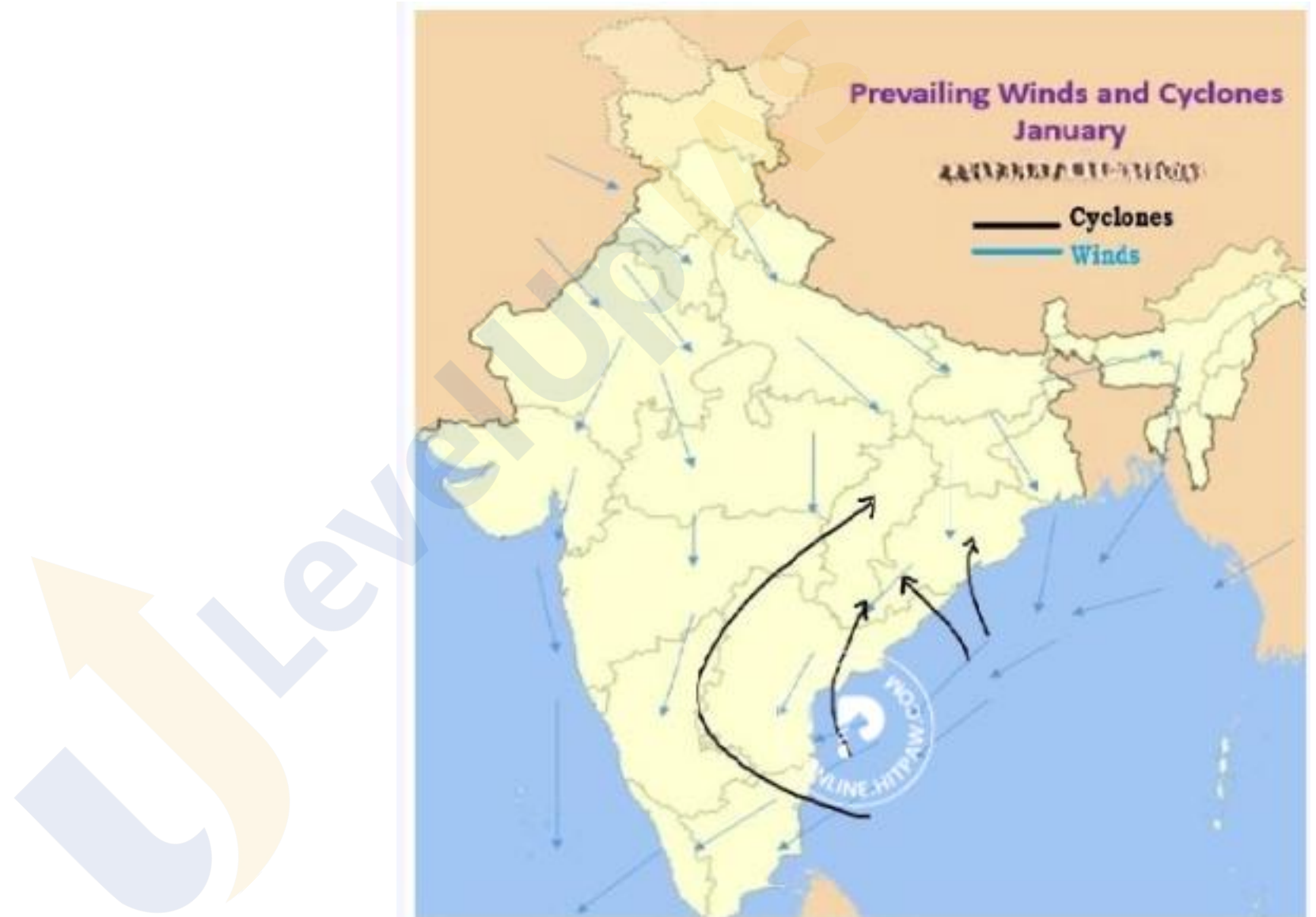
1. Mean daily temp is below 21°C over most parts of northern India.
2. Night temperature is below freezing point in Punjab and Rajasthan.
3. Excessive Cold in North India as
 1. Region is far away from the moderating influence of sea experience continental climate.
 2. Snowfall in Himalayan range creates cold wave situation
 3. Cold winds from Caspian Sea and Turkmenistan bring cold wave along with frost and fog over northwestern parts of India



Climate of India: Winter Season (Nov-Feb)

1. Peninsular region of India does not have any well-defined cold weather season.
2. There is hardly any seasonal change in the distribution pattern of the temperature in coastal areas because of moderating influence of the sea and the proximity to equator.
3. Temp in Thiruvananthapuram in Jan is about 21°C and for June, it is 29.5°C .
4. Temperatures at the hills of Western Ghats remain comparatively low

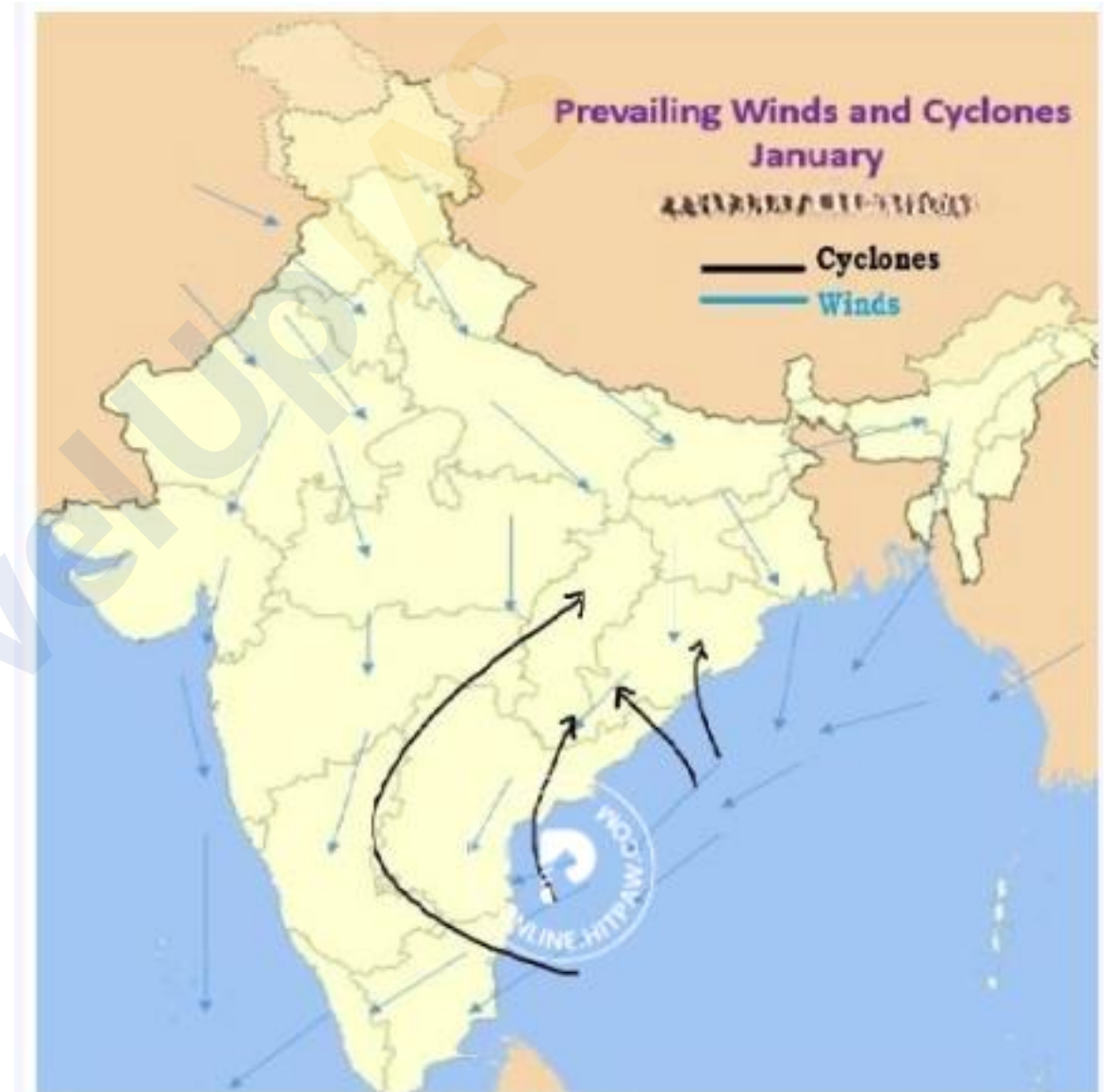




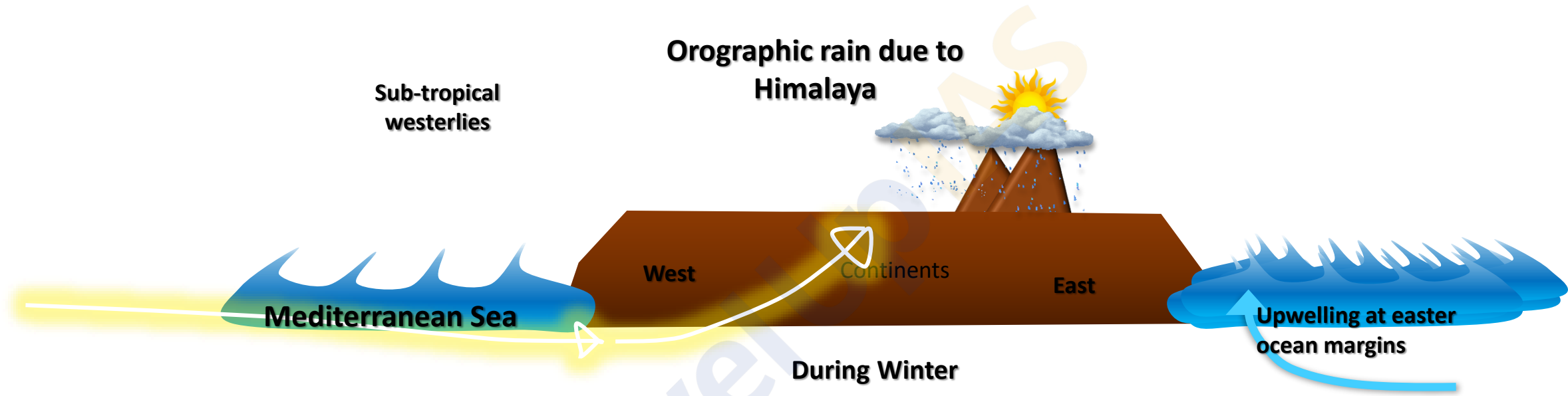
Climate of India: Winter Season (Nov-Jan)

Pressure and Winds :

- By 22nd Dec, the sun shines vertically over the Tropic of Capricorn in the southern hemisphere.
- India has high pressure conditions over the northern plain (1019 mb)
- In south India, the air pressure is slightly lower (1013 mb)
- Winds start blowing from north-western high pressure zone to low air pressure zone over Indian Ocean in south.
- Due to low pressure gradient, winds blow with low velocity of about 3-5 km per/hour
- During the winters, the weather in India is pleasant.

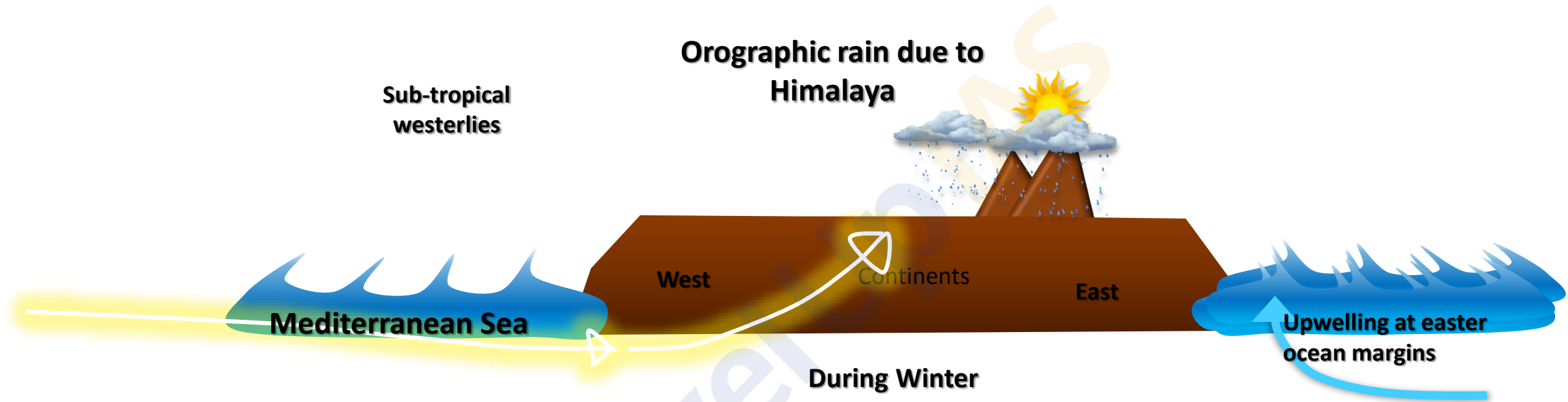


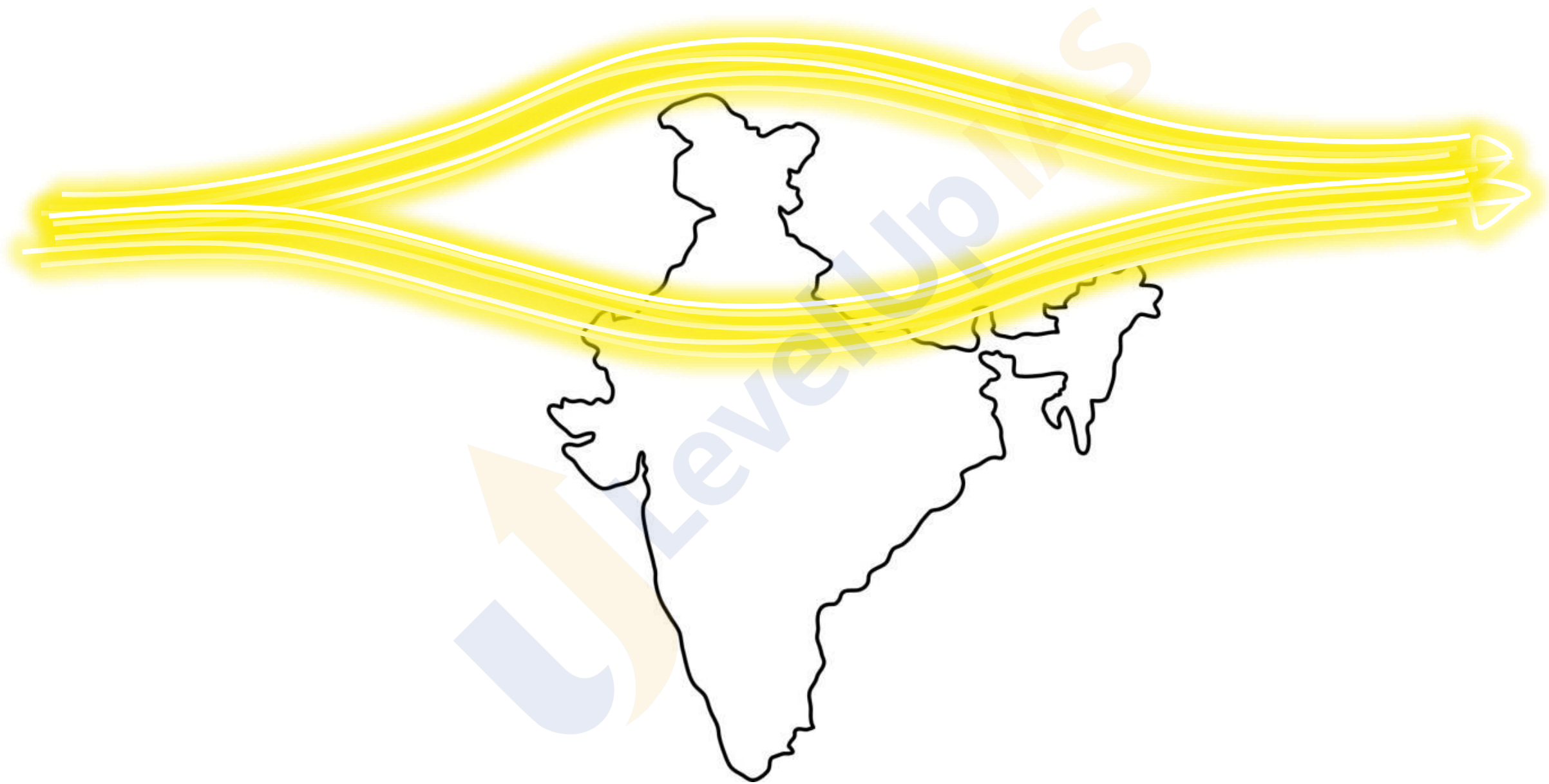
Winter rain

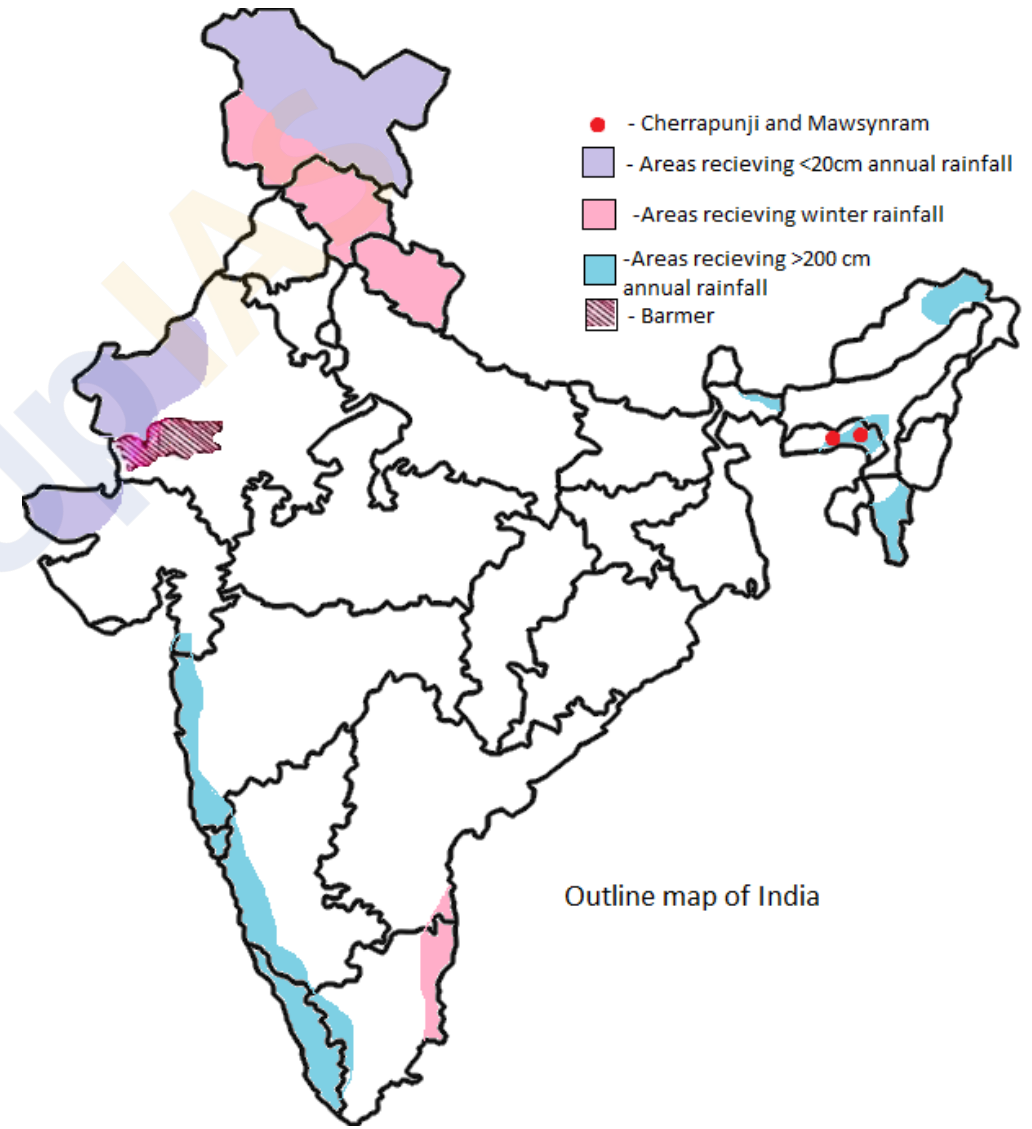


- The pleasant weather gets disturbed by shallow cyclonic depressions originating over the east Mediterranean Sea and travelling eastwards across West Asia, Iran, Afghanistan and Pakistan before they reach the north-western parts of India.
- On their way, the moisture content gets augmented from the Caspian Sea in the north and the Persian Gulf in the south.

Winter rain







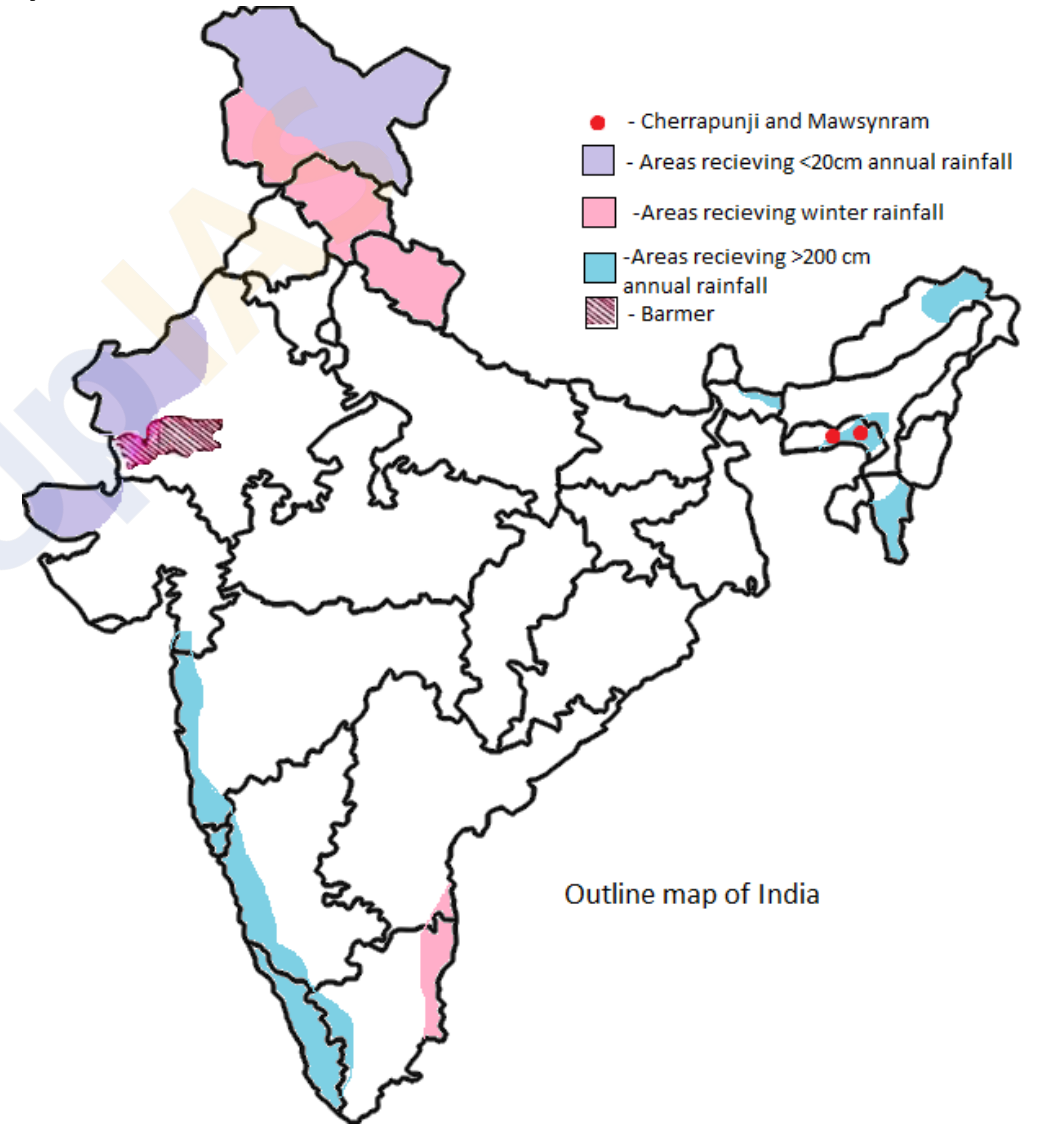
Outline map of India

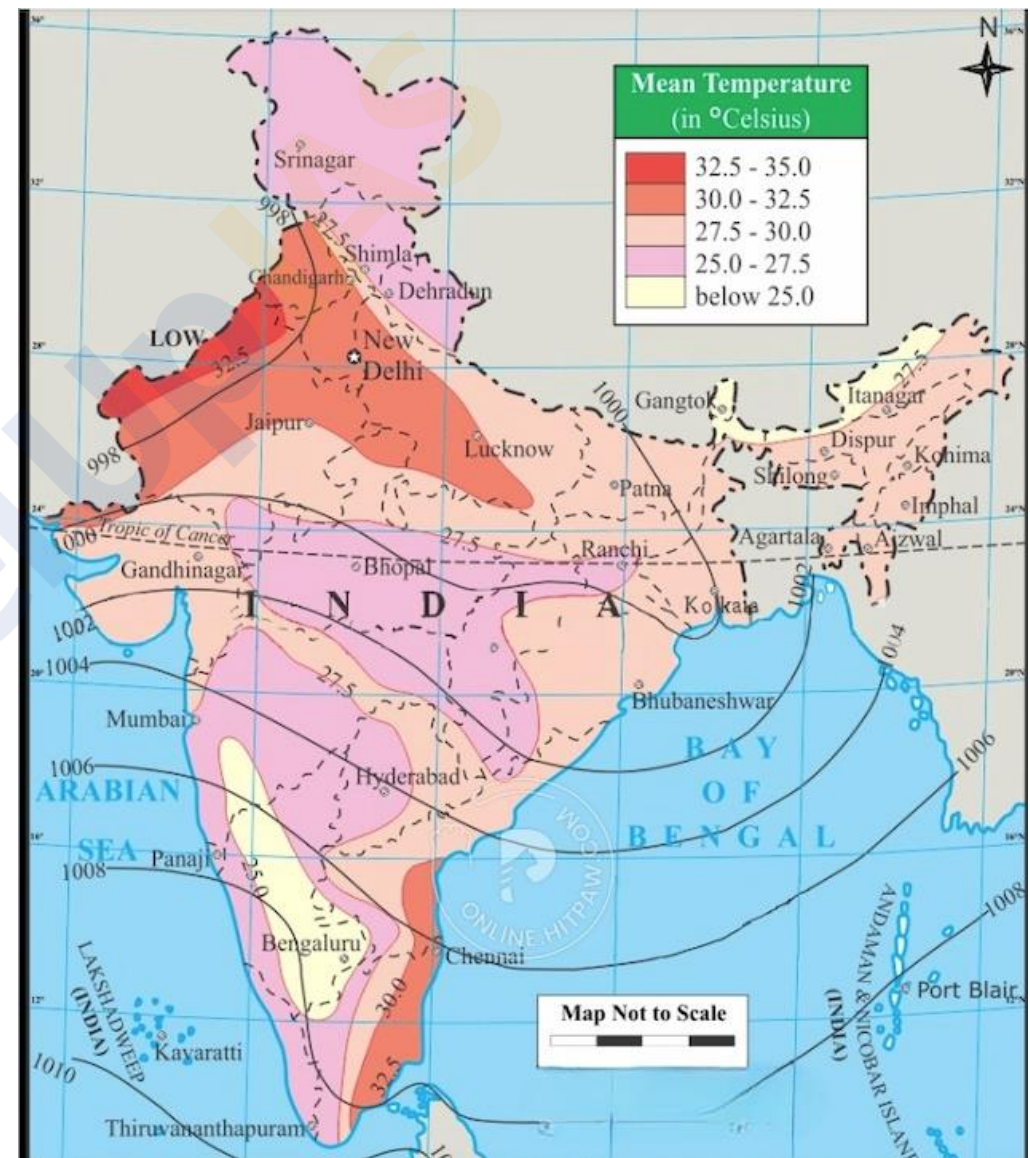
Rainfall in Winter Season (Nov-Jan)

1. Winter monsoons do not cause rainfall in major parts of India as they move from land to the sea. Winds have little humidity and also due to anti cyclonic circulation on land, the possibility of rainfall reduces.

Some places that receive rainfall are

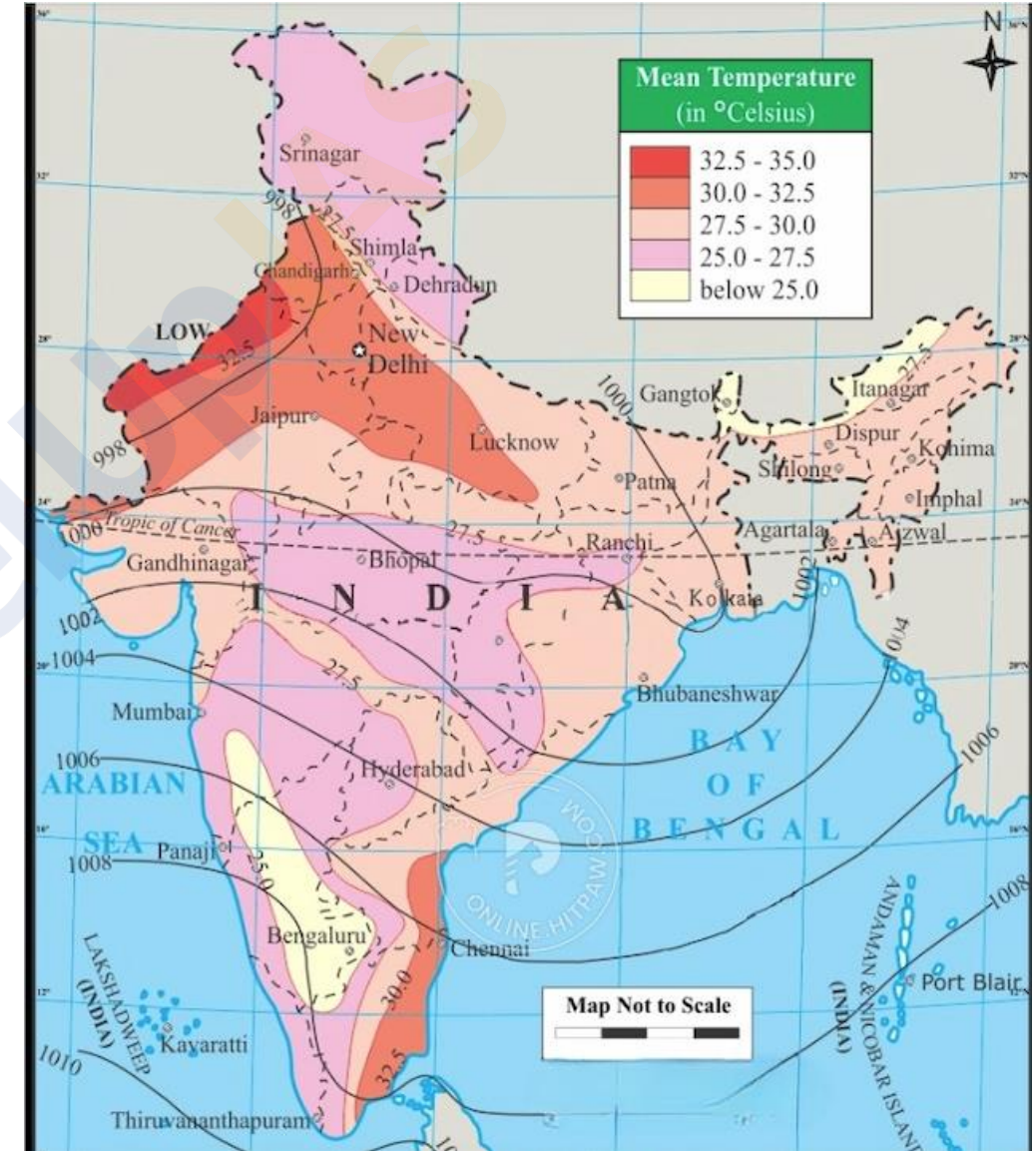
1. North-western India due to temperate cyclones from Med sea. This rain is beneficial for rabi crops
2. Central parts of India and northern parts Peninsula
3. Arunachal Pradesh and Assam in north-east India
4. Northeast monsoon while crossing over Bay of Bengal, picks up moisture and causes rainfall over Tamil Nadu coast, southern Andhra Pradesh an sometime upto southeast Karnataka and southeast Kerala.





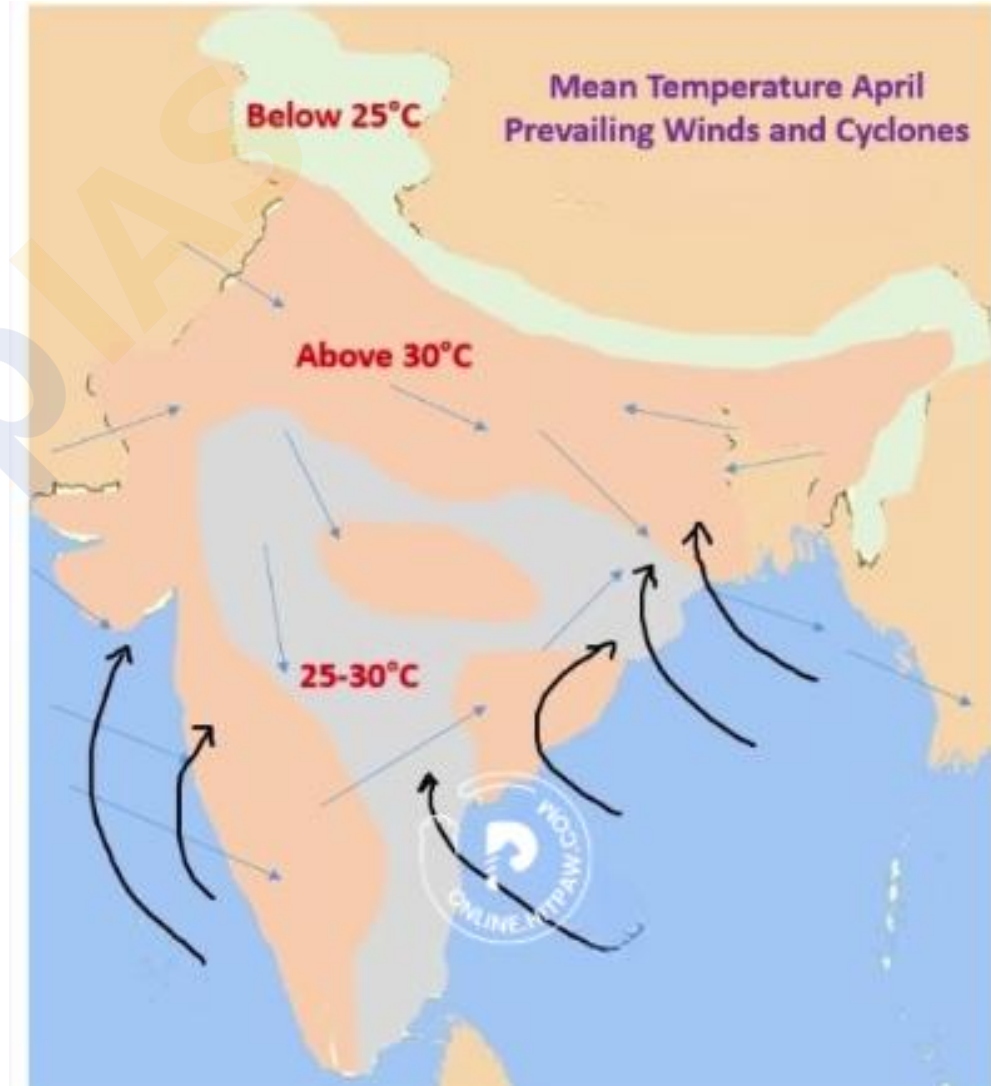
Summer Season (15th March- 15th June)

- **Temperature:** Due to Apparent movement of sun from equator to 23.5 °N from March to June, temperatures start rising in north India. Temperature can rise and become 40 °C
- Hot weather season in **south India is mild and not so intense** as found in north due to moderating effect of ocean with temperature being around 26°C and 32°C. Due to altitude, the temperatures in the hills of Western Ghats remain below 25°C.
- In **coastal regions**, isotherms are parallel to coast and run north to south which confirms that temperature does not decrease from north to south rather it increases from coast to interior.



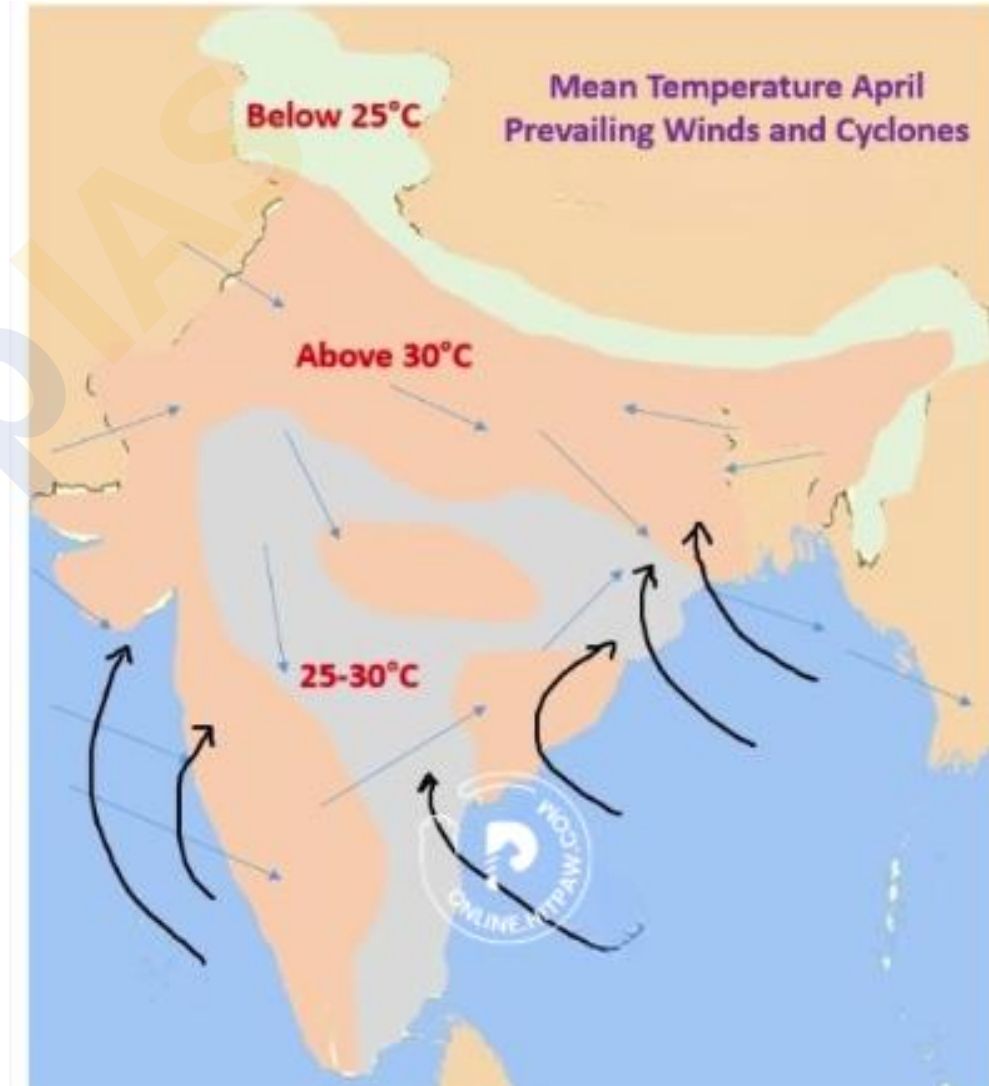


LevelUp



Summer Season (15th March- 15th June)

- **Pressure and Winds:** Excessive heat in summer creates LP zone from Thar desert to Chota nagpur plateau.
- **ITCZ moves northwards** and is centred at 25°N in July. Which **attracts surface of winds:** south-westerly on the west coast and along Bay of Bengal. They are easterly over Bengal and Bihar
- Along ITCZ, dry and **hot winds known as 'Loo'**, blow in afternoon which continue in midnight
- Dust storms are common **in PJ, HR, Raj and UP**. Storms bring welcome respite from heat as they bring light rains and pleasant cool breeze.
- Local storms of great intensity associated with violent winds, torrential rains and hailstorms are common

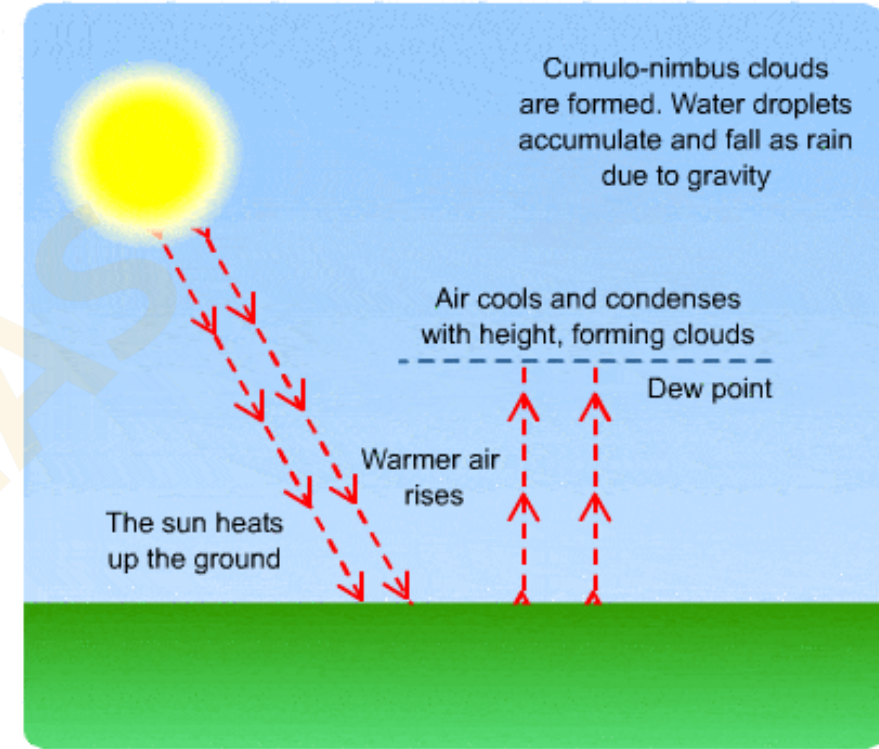




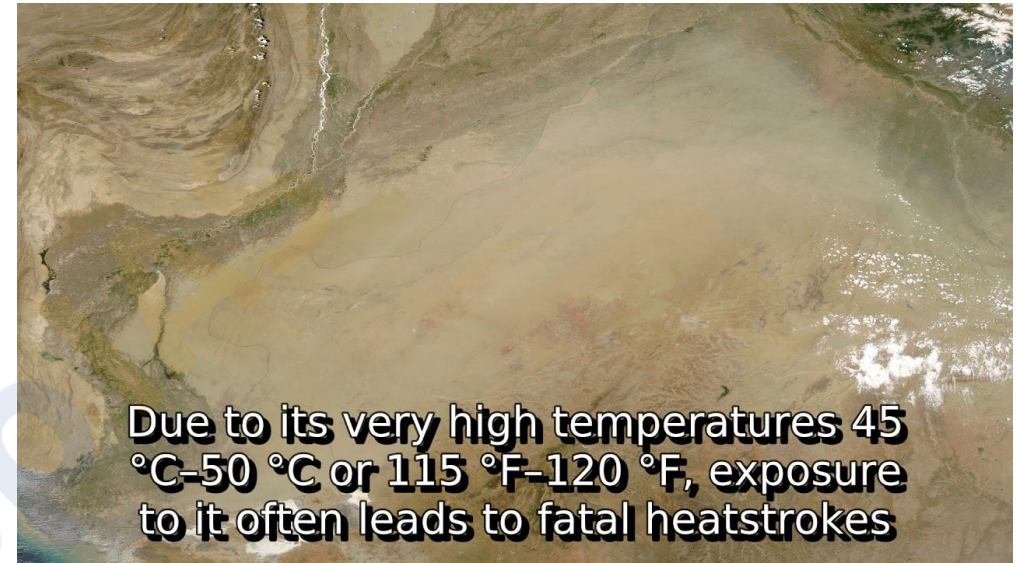
LOW PRESSURE ZONE

Some Famous Local Storms of Hot Weather Season

- (i) *Mango Shower* : Towards the end of summer, there are pre-monsoon showers which are a common phenomena in Kerala and coastal areas of Karnataka. Locally, they are known as mango showers since they help in the early ripening of mangoes.
- (ii) *Blossom Shower* : With this shower, coffee flowers blossom in Kerala and nearby areas.
- (iii) *Nor Westers* : These are dreaded evening thunderstorms in Bengal and Assam. Their notorious nature can be understood from the local nomenclature of '*Kalbaisakhi*', a calamity of the month of *Baisakh*. These showers are useful for tea, jute and rice cultivation. In Assam, these storms are known as "*Bardoisila*".
- (iv) *Loo* : Hot, dry and oppressing winds blowing in the Northern plains from Punjab to Bihar with higher intensity between Delhi and Patna.







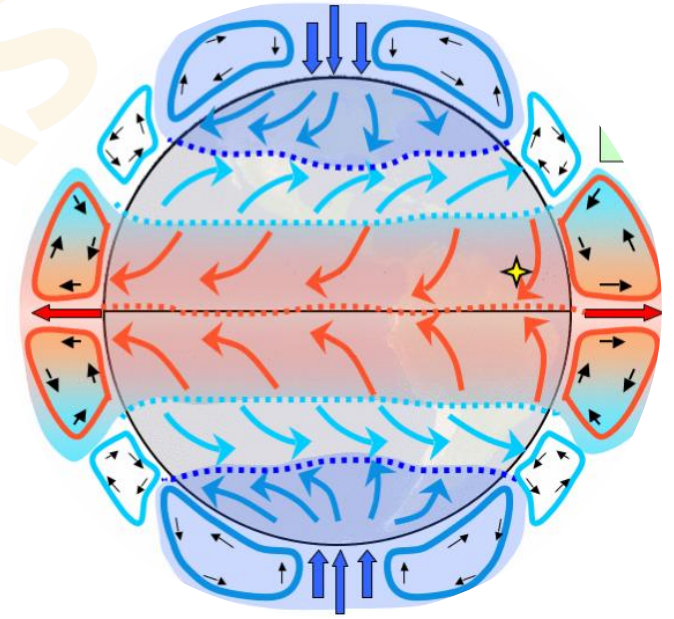
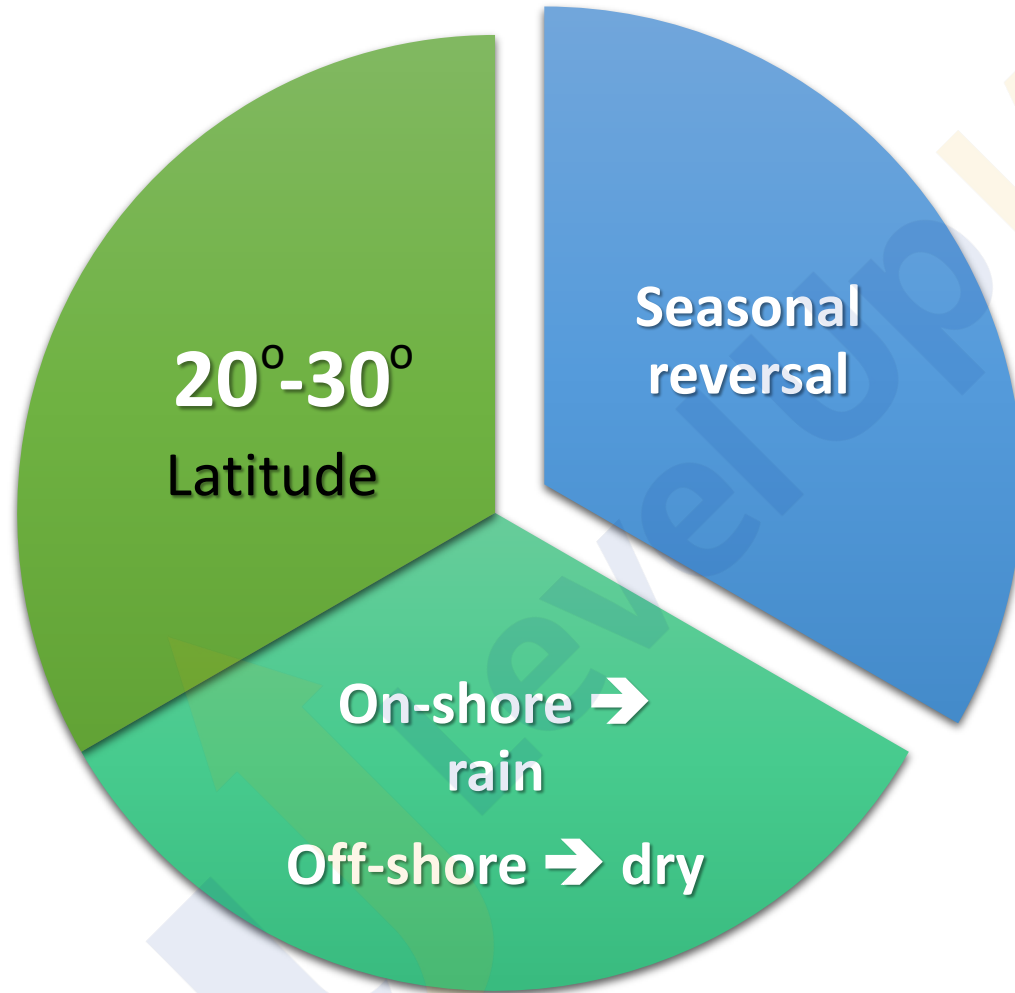
Due to its very high temperatures 45 °C-50 °C or 115 °F-120 °F, exposure to it often leads to fatal heatstrokes



Monsoon



Characteristic



2014

The seasonal reversal of wind is characteristic feature of?

- a) Equatorial Climate
- b) Mediterranean Climate
- c) Monsoon Climate
- d) All of the above



2012

C o n s i d e r t h e f o l l o w i n g s t a t e m e n t s :

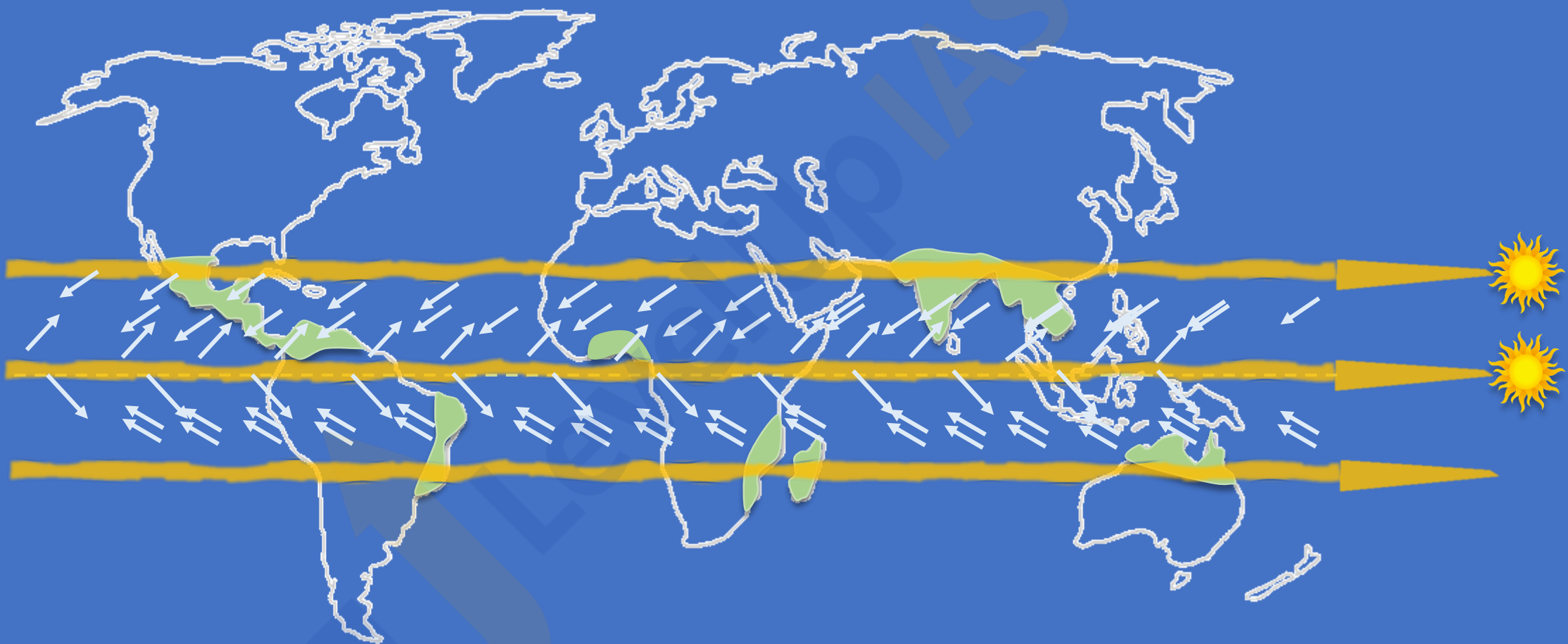
1. T h e d u r a t i o n o f t h e m o n s o o n
d e c r e a s e s f r o m s o u t h e r n I n d i a t o
n o r t h e r n I n d i a .
2. T h e a m o u n t o f a n n u a l r a i n f a l l i n t h e
n o r t h e r n p l a i n s o f I n d i a d e c r e a s e s
f r o m e a s t t o w e s t .

**W h i c h o f t h e s t a t e m e n t s g i v e n a b o v e
i s / a r e c o r r e c t ?**

- (a) 1 o n l y
- (b) 2 o n l y
- (c) B o t h 1 a n d 2
- (d) N e i t h e r 1 n o r 2



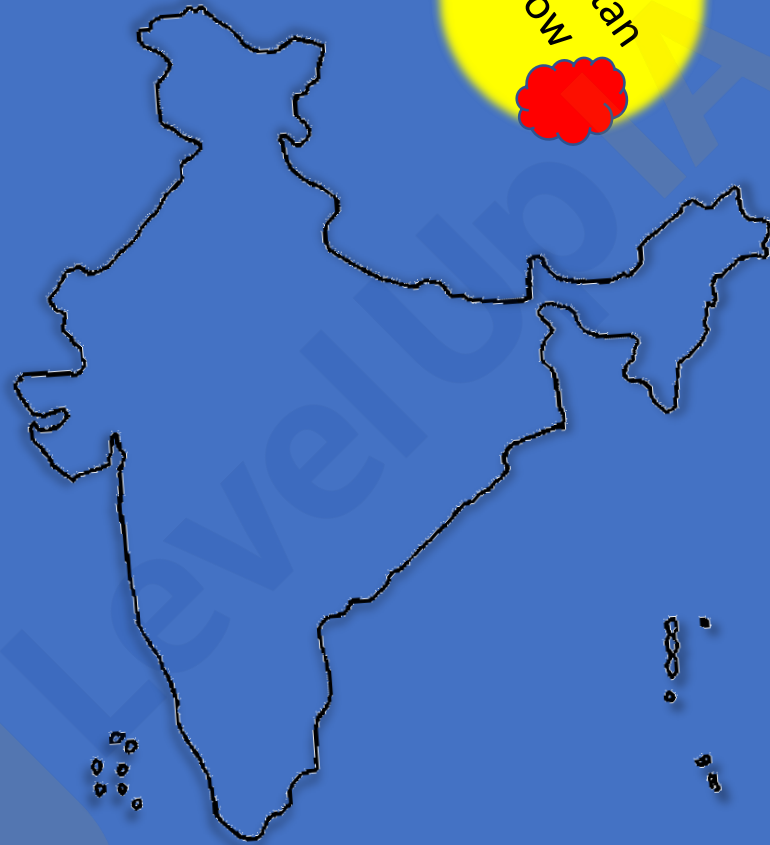
Arabic word 'Mausim'

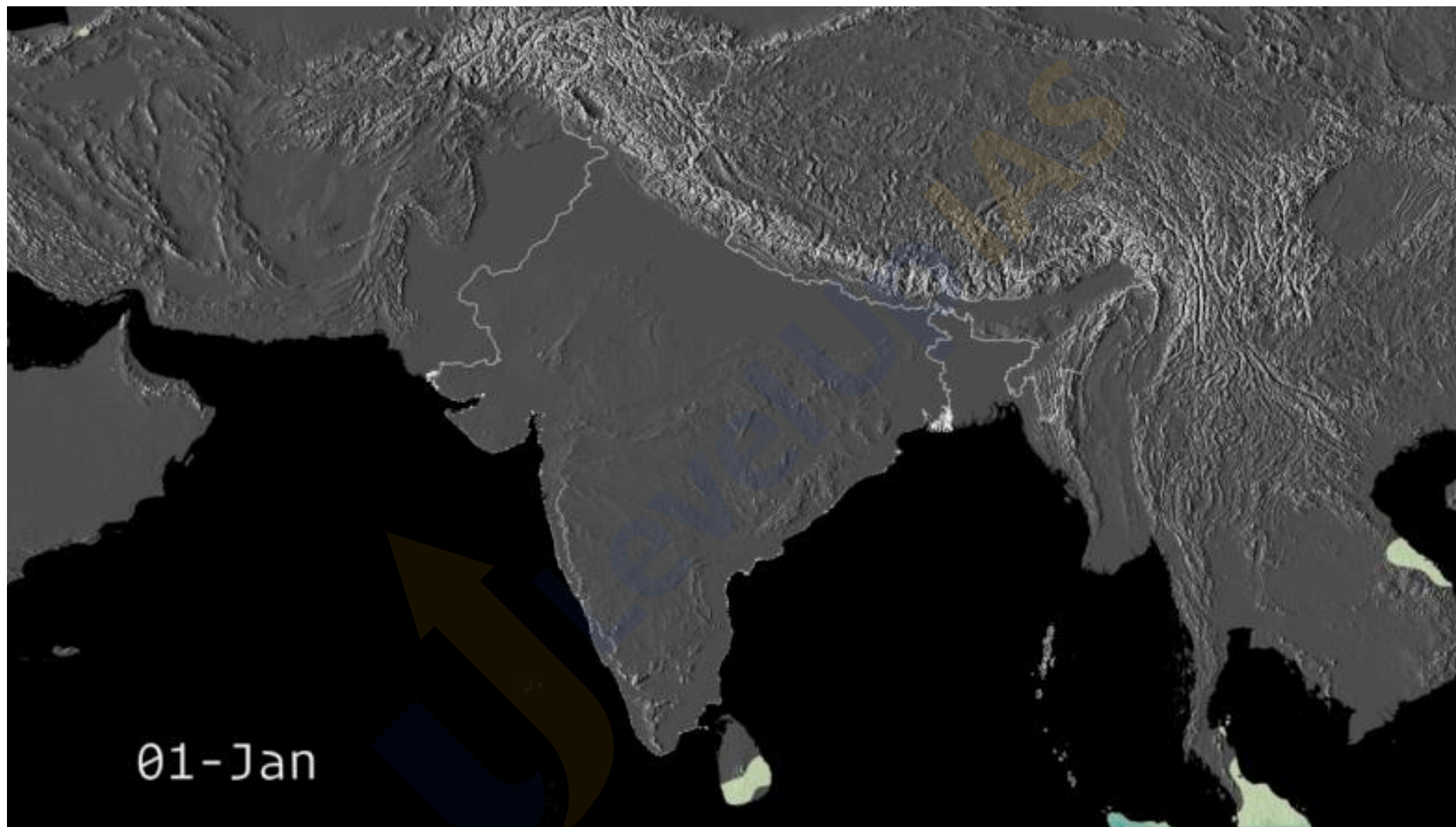


Tropopause

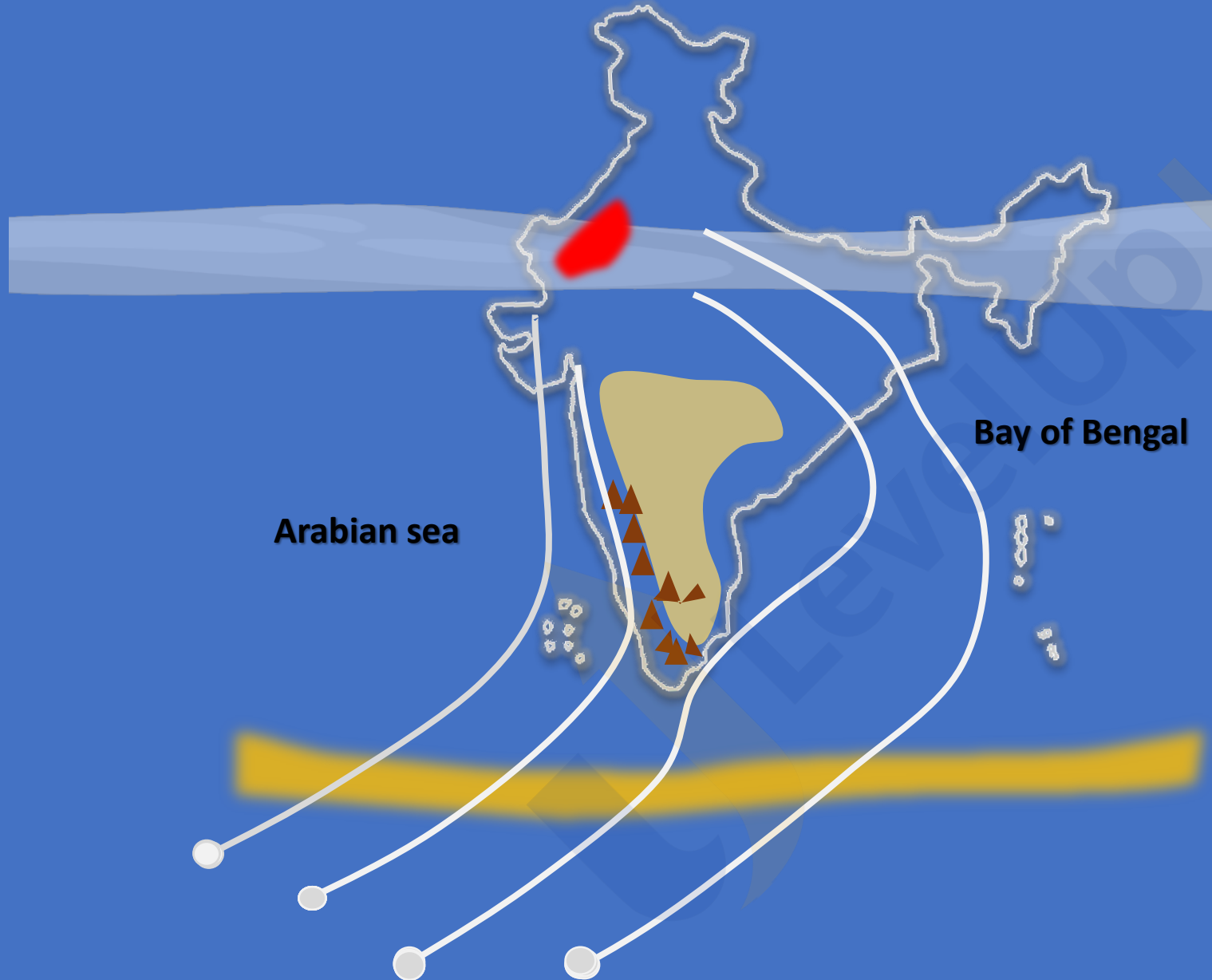
Tibetan
low

Mascarenes
high





Arrival of monsoon



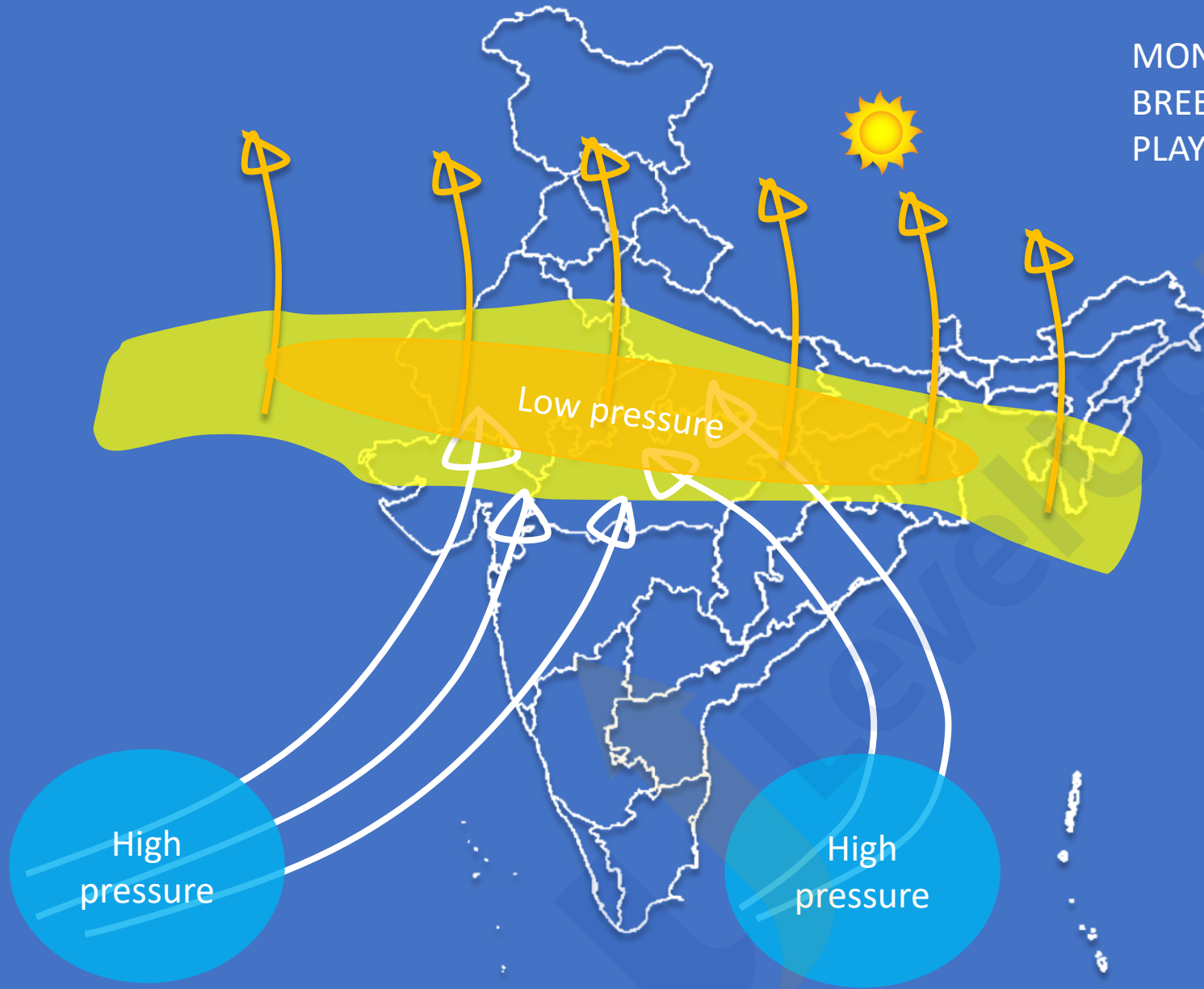
Northward shift of Sun

Pressure belts shift north ward

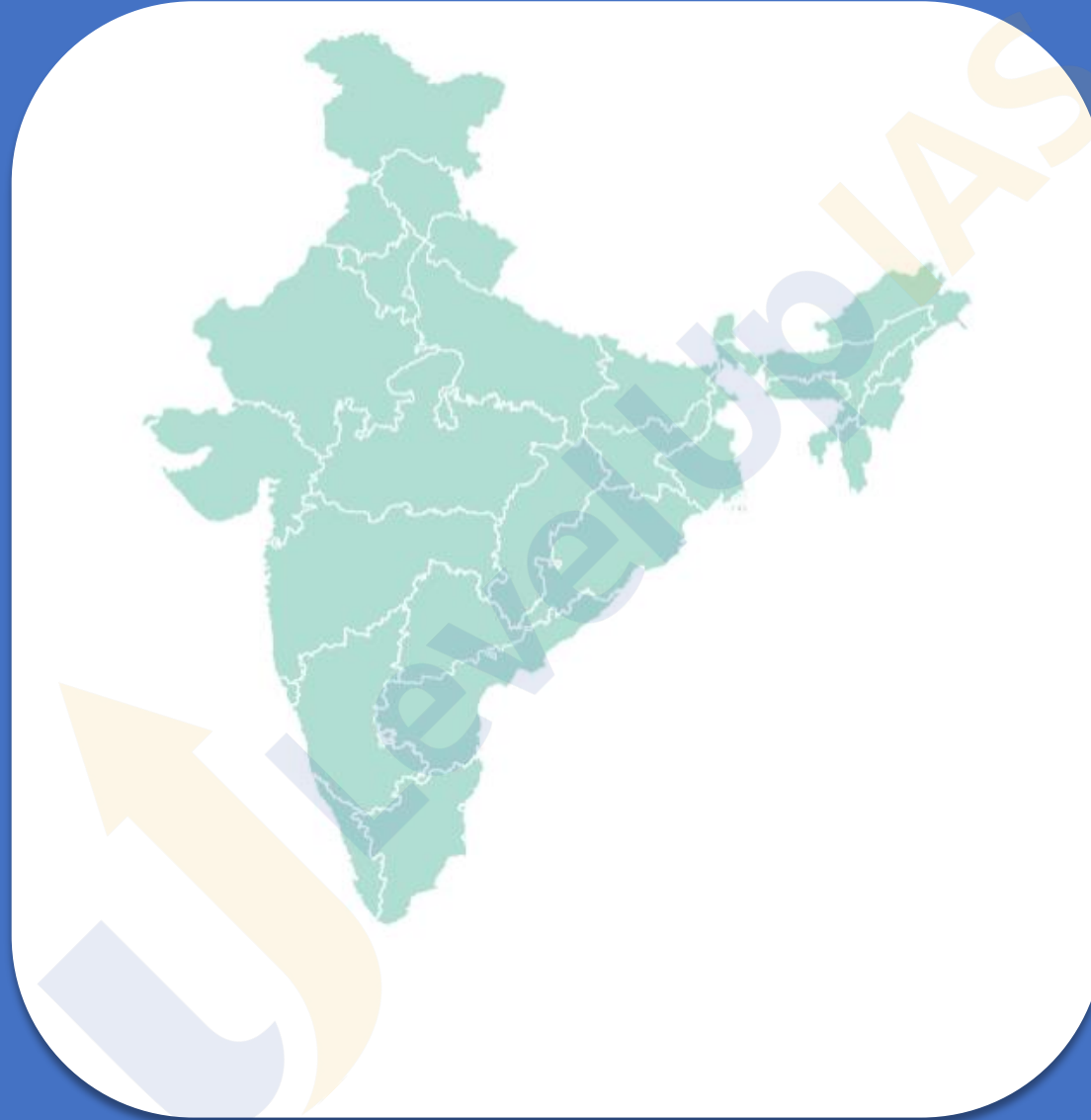
In India, ITCZ gets placed over the northern plain

Trade winds get attracted towards the subcontinent causing the Monsoon "Burst".

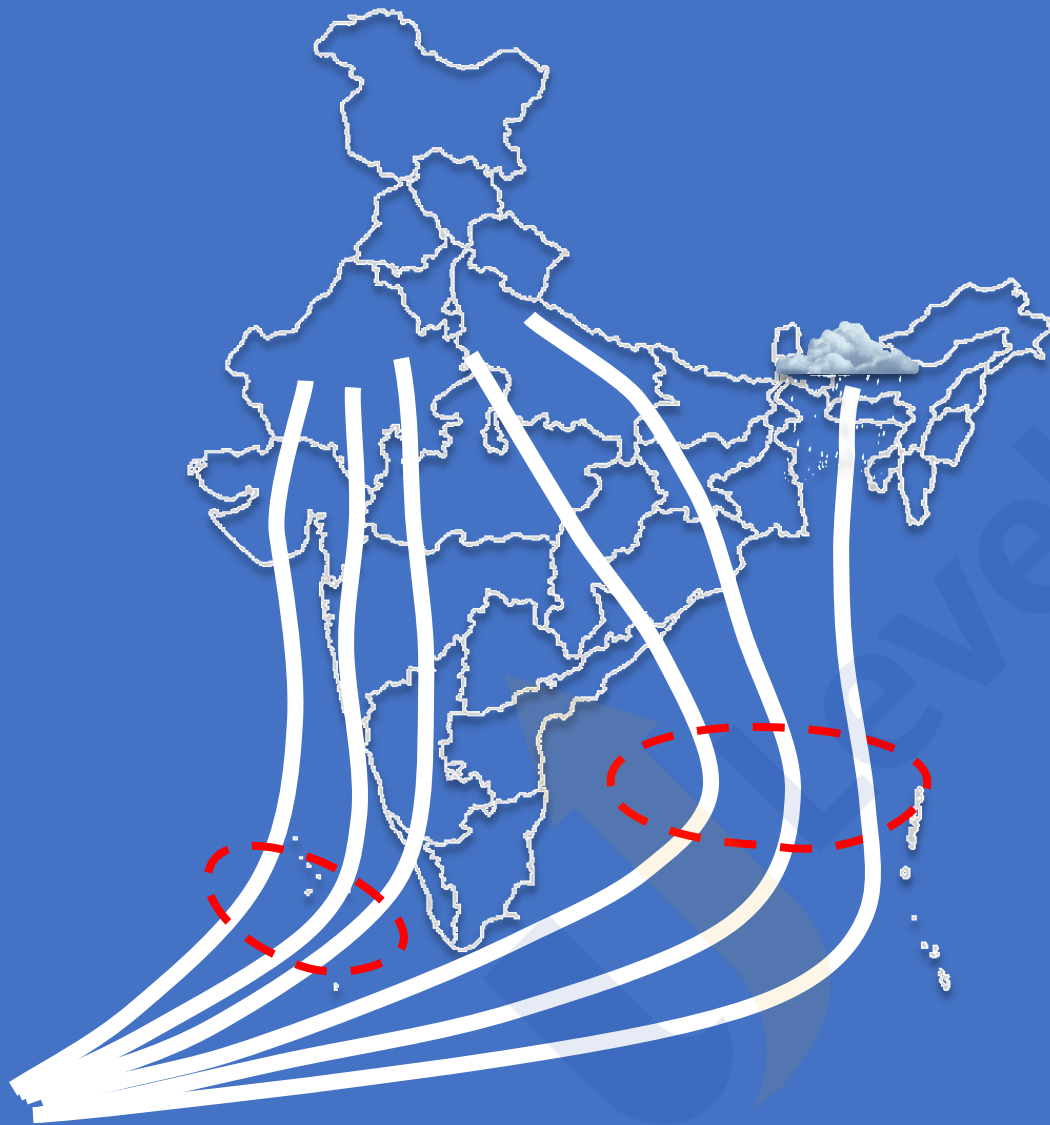
MONSOON ARE NOT SIMPLY LAND AND SURFACE BREEZE: SUB TROPICAL WESTERLY JETSTREA ASLO PLAYS AN IMP ROLE:



Progress of monsoon



Progress of monsoon



arrives at the southern tip of the Indian peninsula

Causes sudden Burst of rain

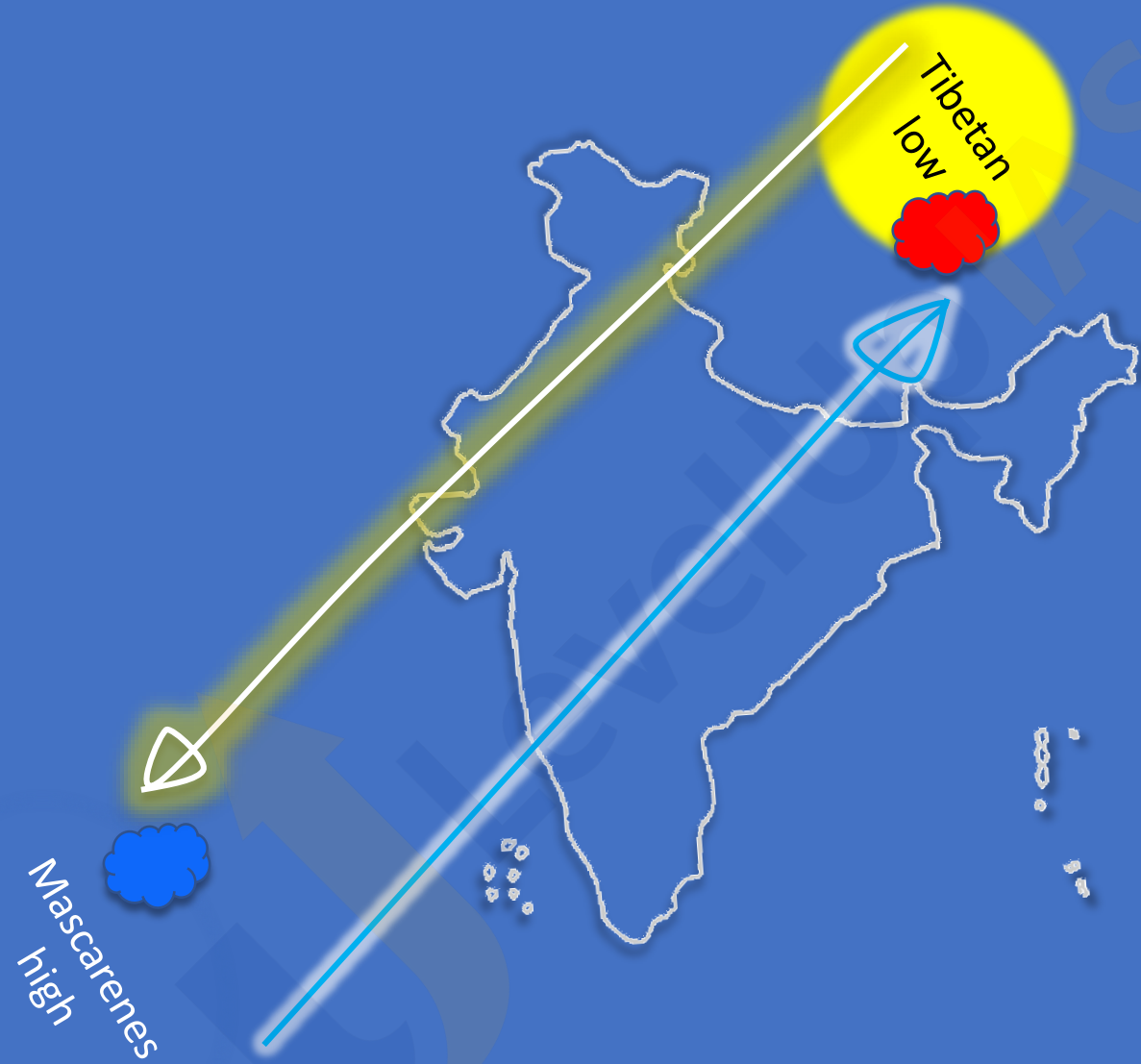
The winds gets bifurcated into two branches

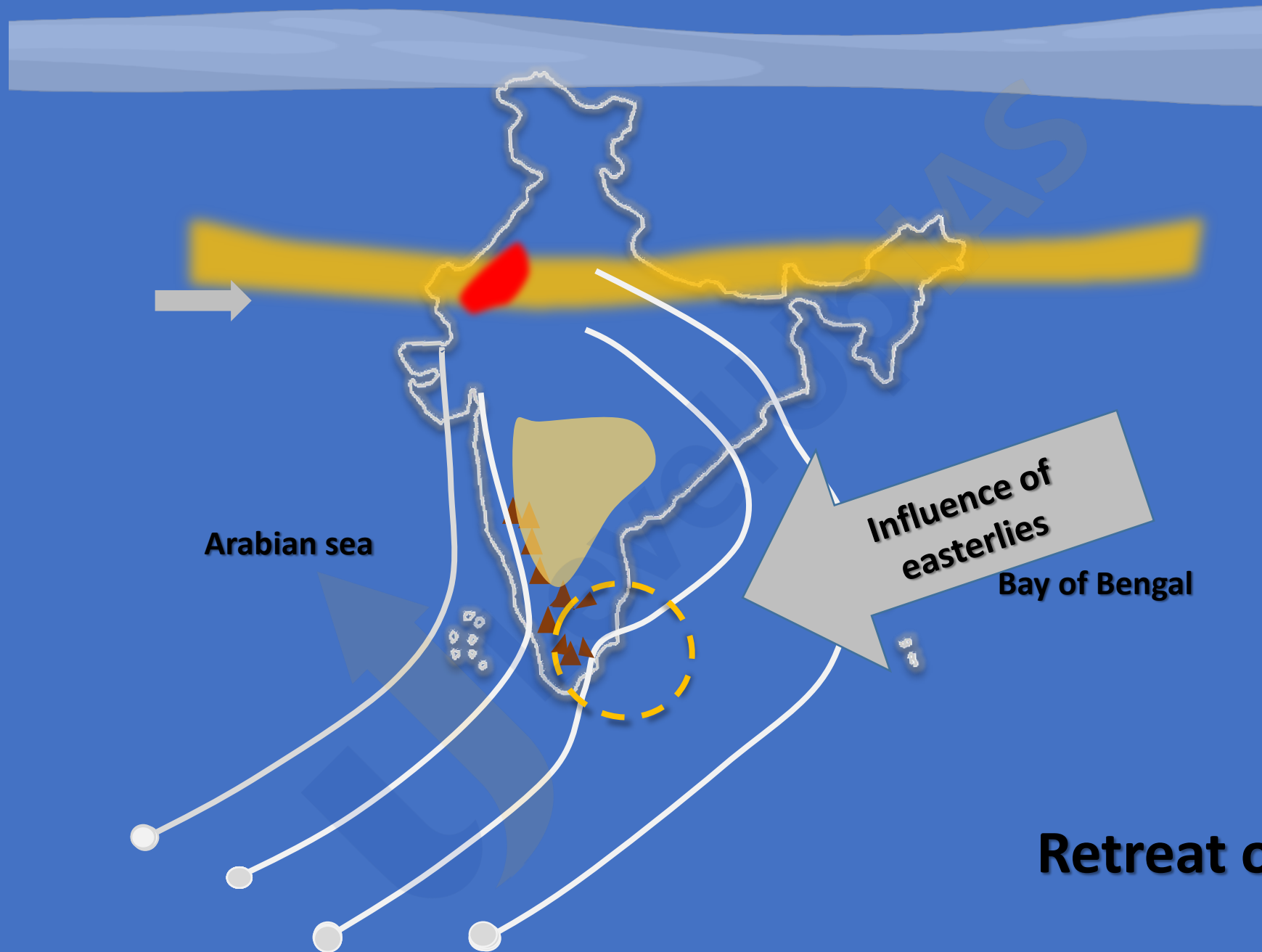
Arabian sea Branche and Bay of Bengal Branche

Progress of monsoon



Tropopause





Influence of
easterlies

Bay of Bengal

Arabian sea

Retreat of monsoon

Retreat of monsoon

Apparent
movement of sun
towards south

Starts from
north to south

Rapid in
comparison to
progress

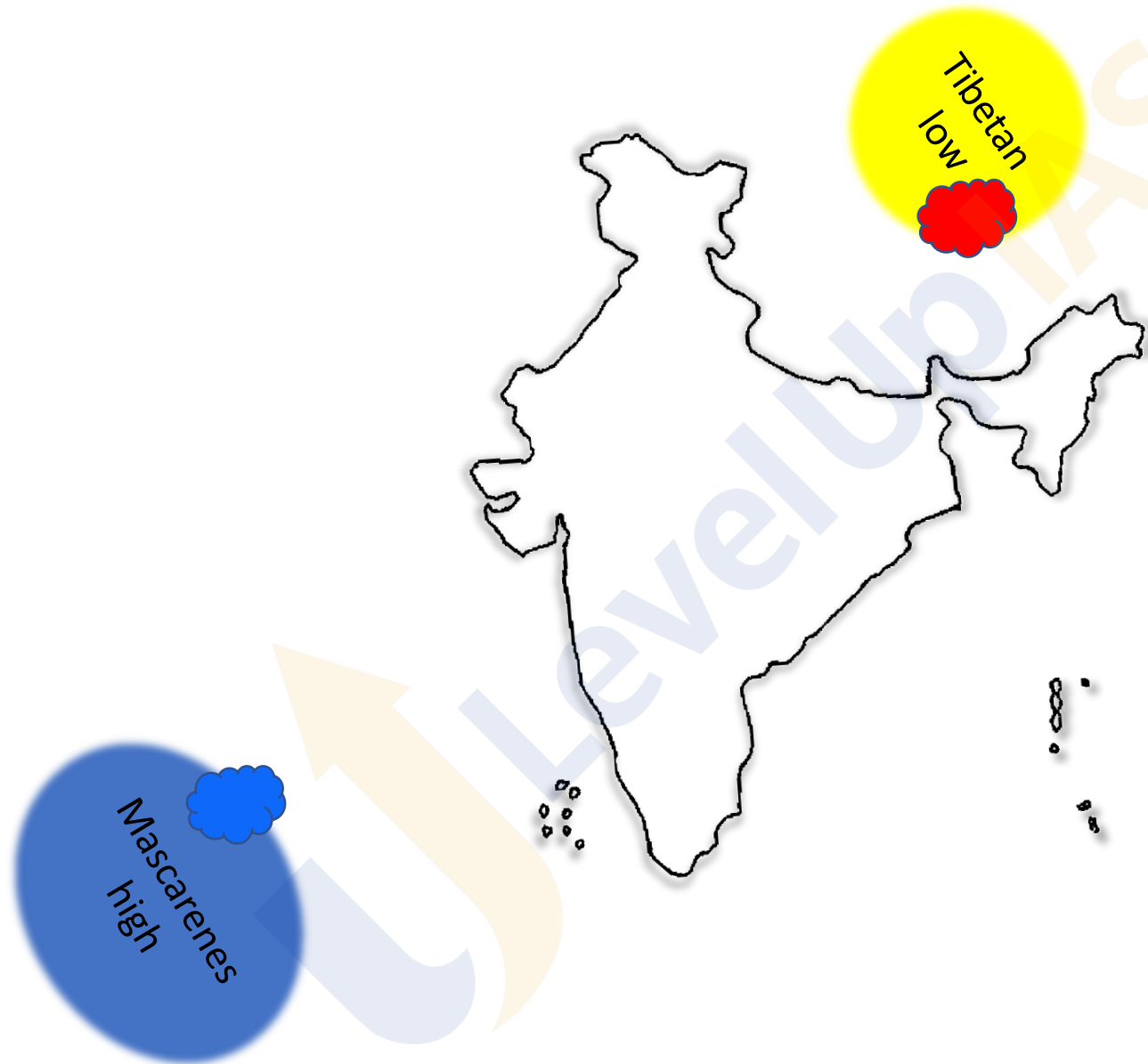
October heat

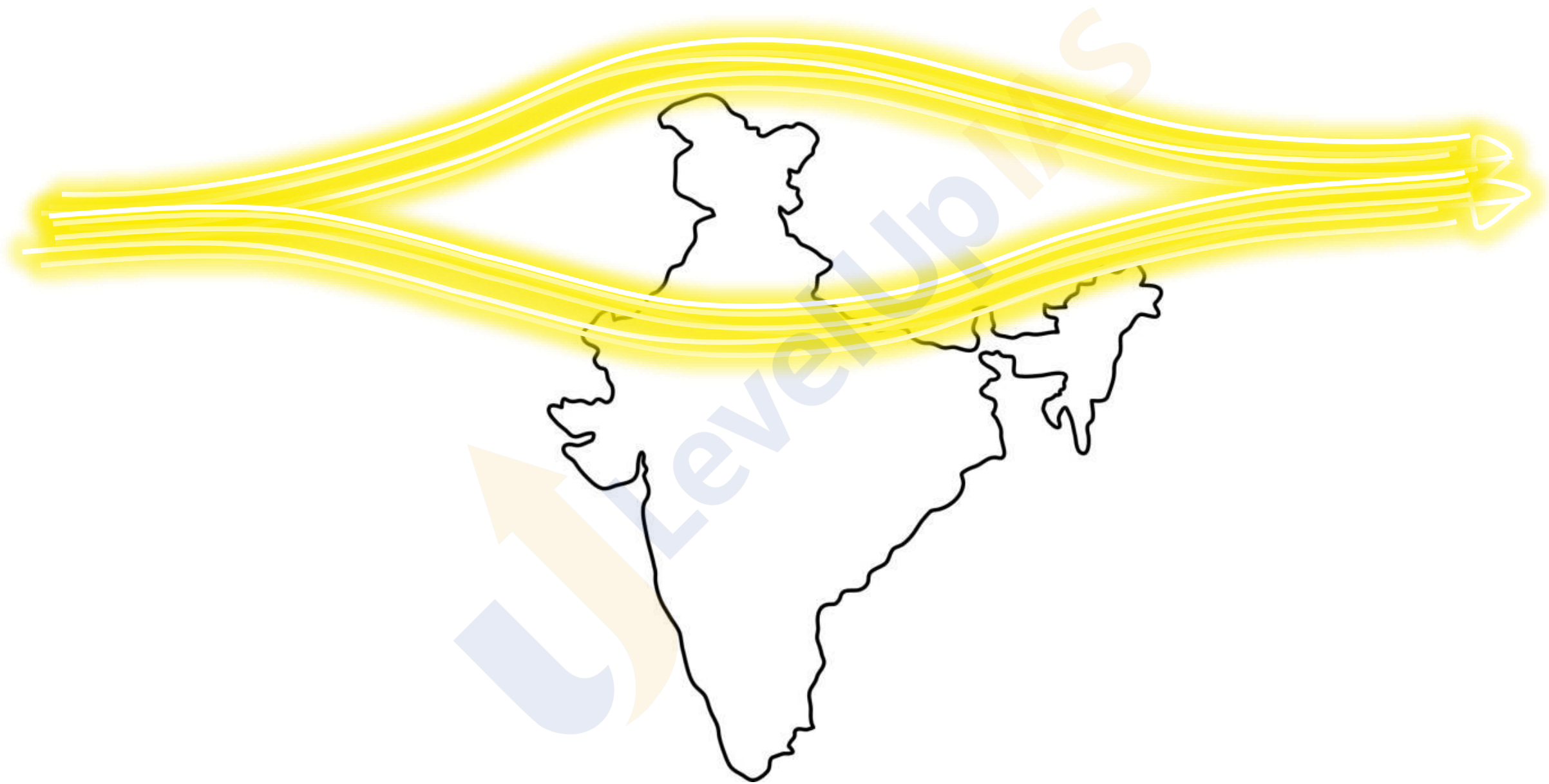
Tropical
cyclone

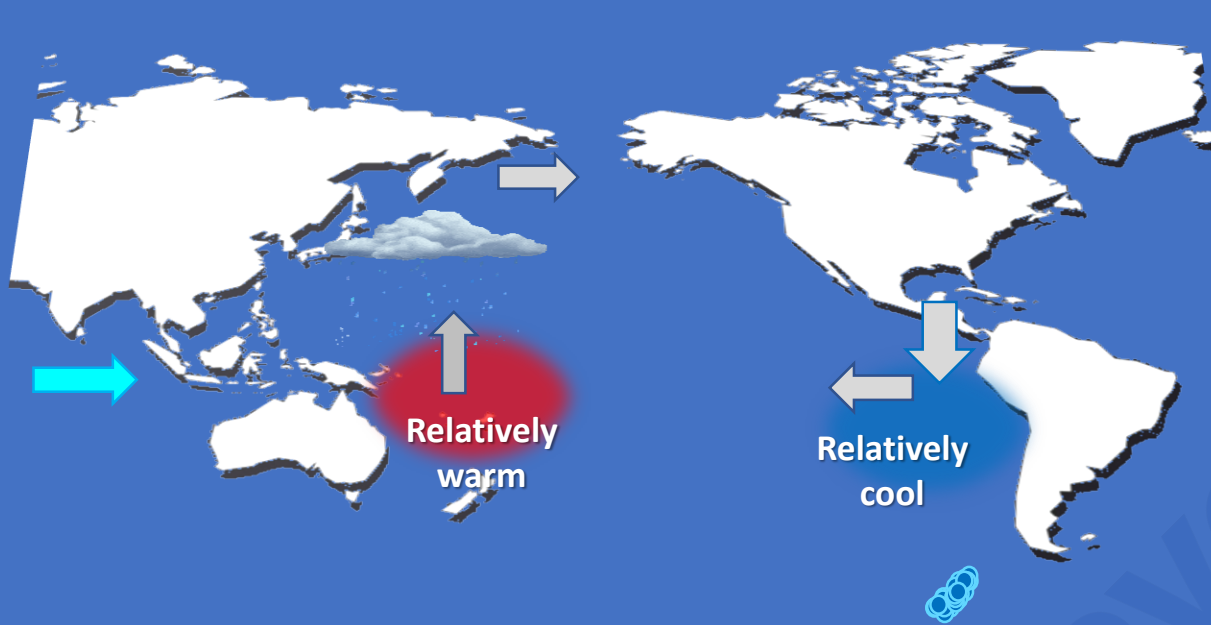
Rain at
southeast
coast

- clear skies
- Day temperatures are high
- Land is still moist.
- High temperature and humidity, weather oppressive during day

Tropopause

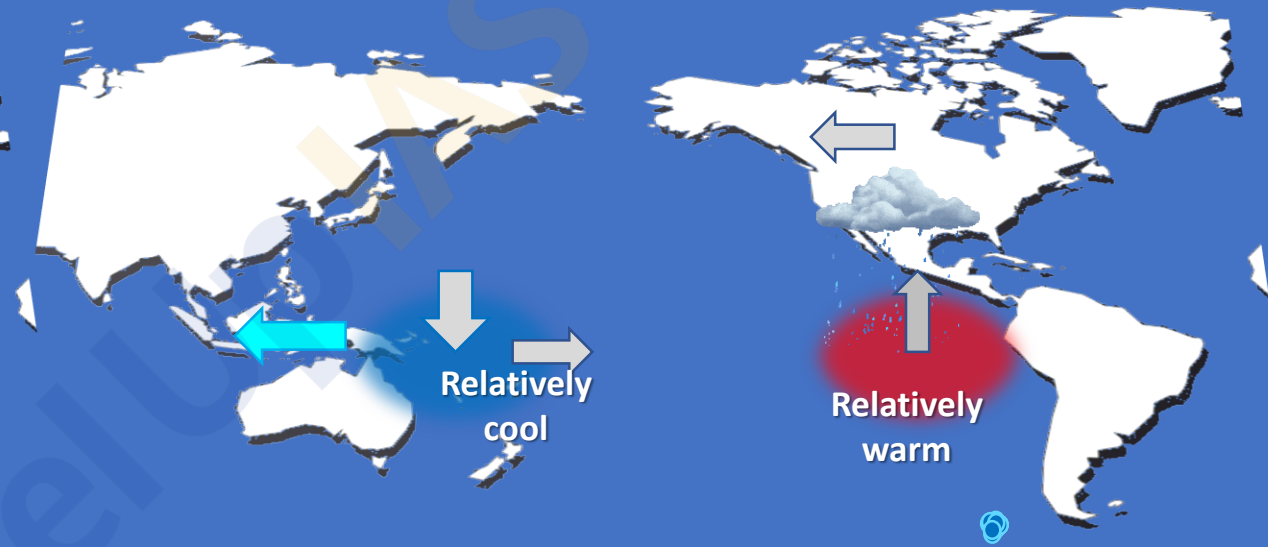






Normal / La nina

Stronger trade winds pushes warm pacific water



El Nino

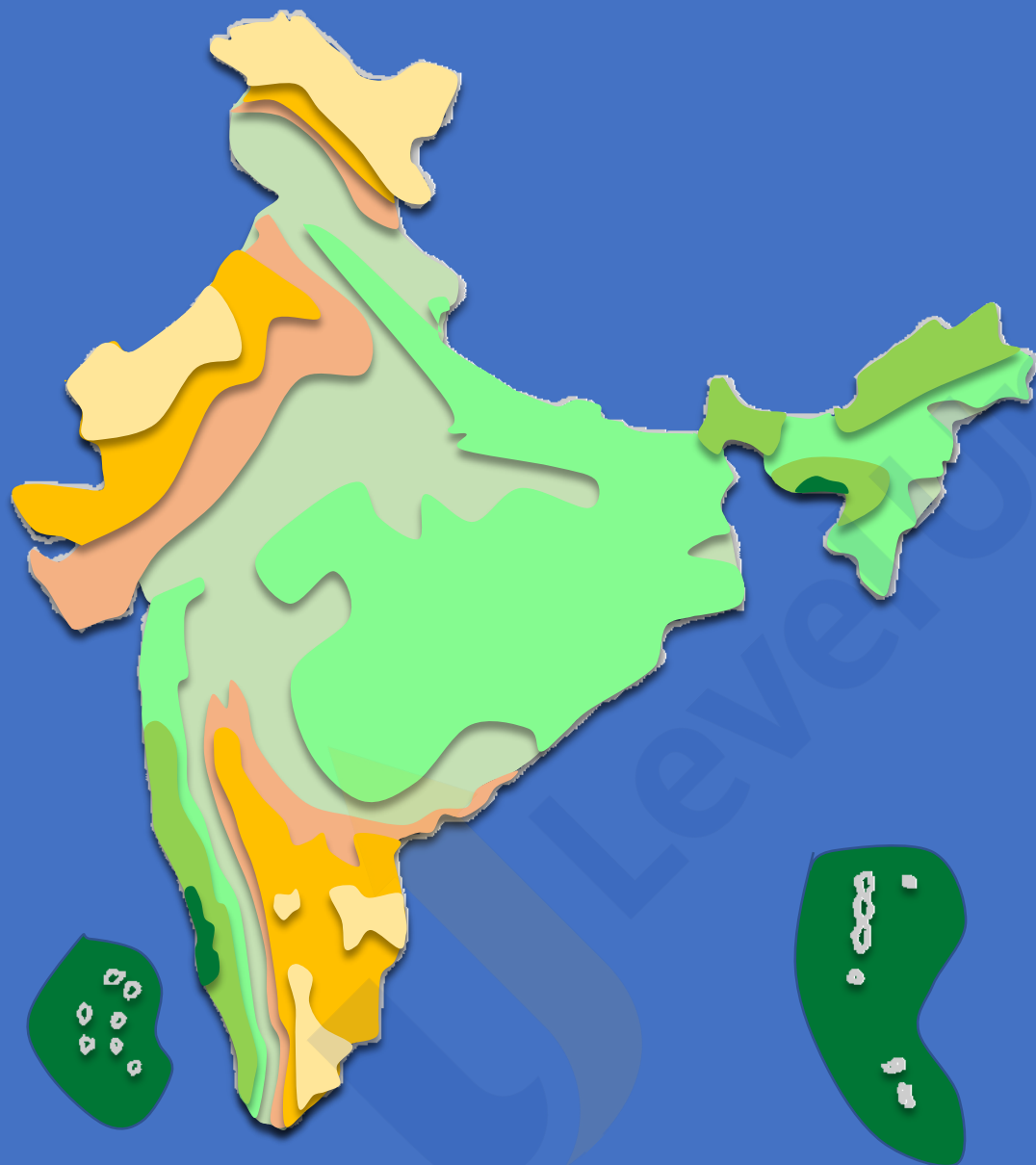
Weakened trade winds unable to push warm pacific water



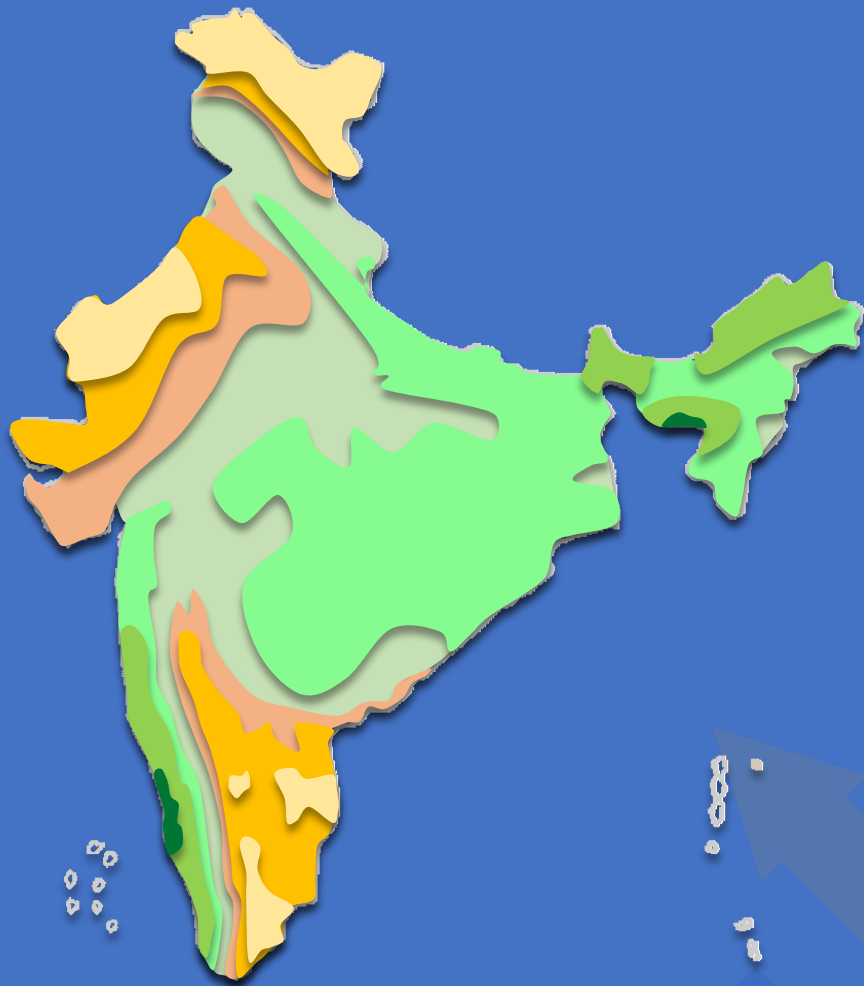
Positive IOD



Negative IOD



Rainfall in Cm	
	> 400cm
	200-400cm
	100-200cm
	60-100cm
	40-60cm
	20-60cm
	0-20cm



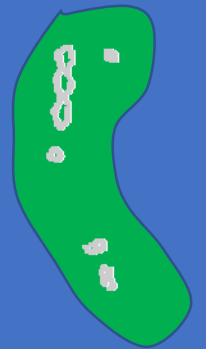
**Tropical Thorn
forest**

Montane forest

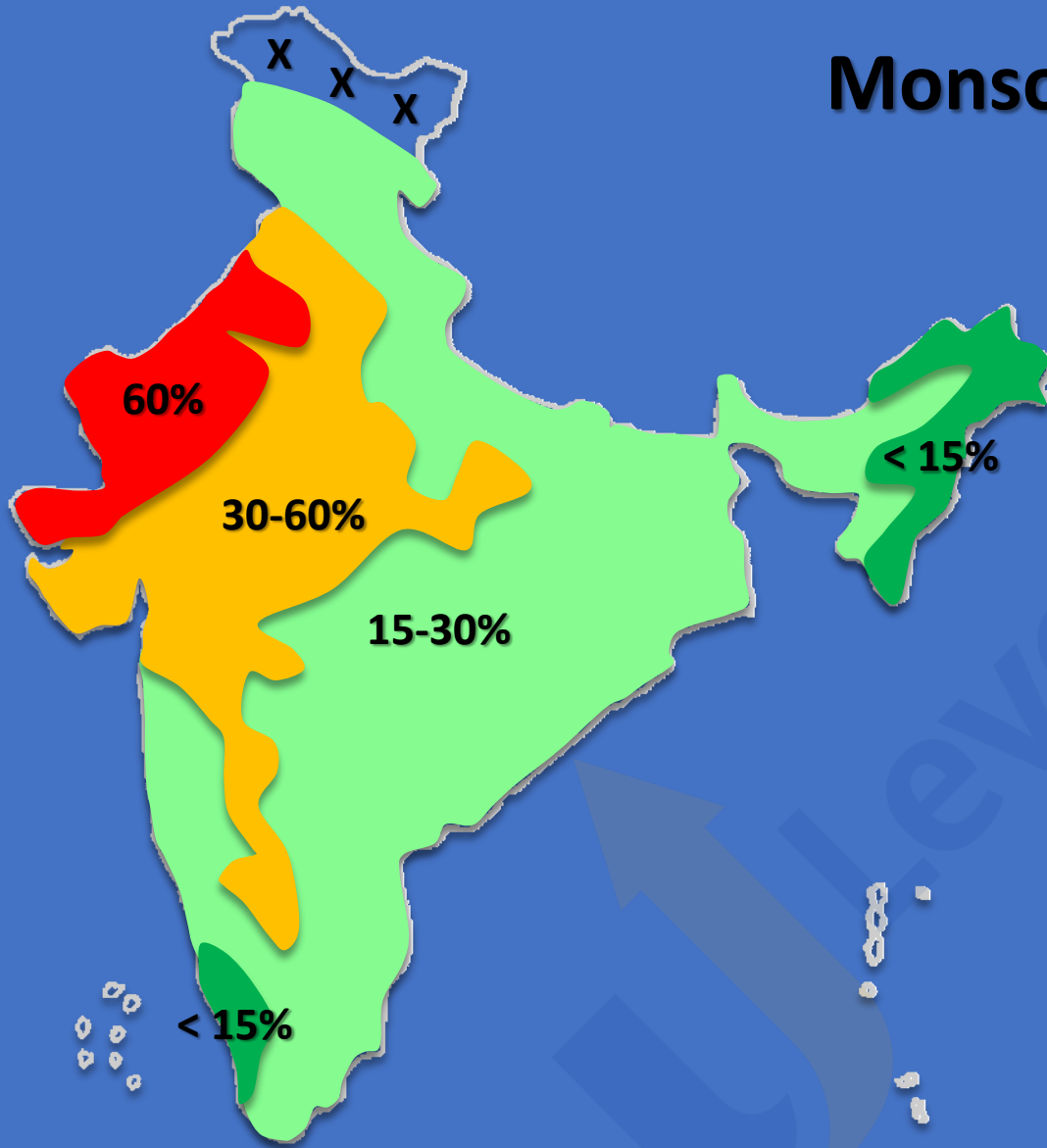
**Tropical Deciduous
forest**

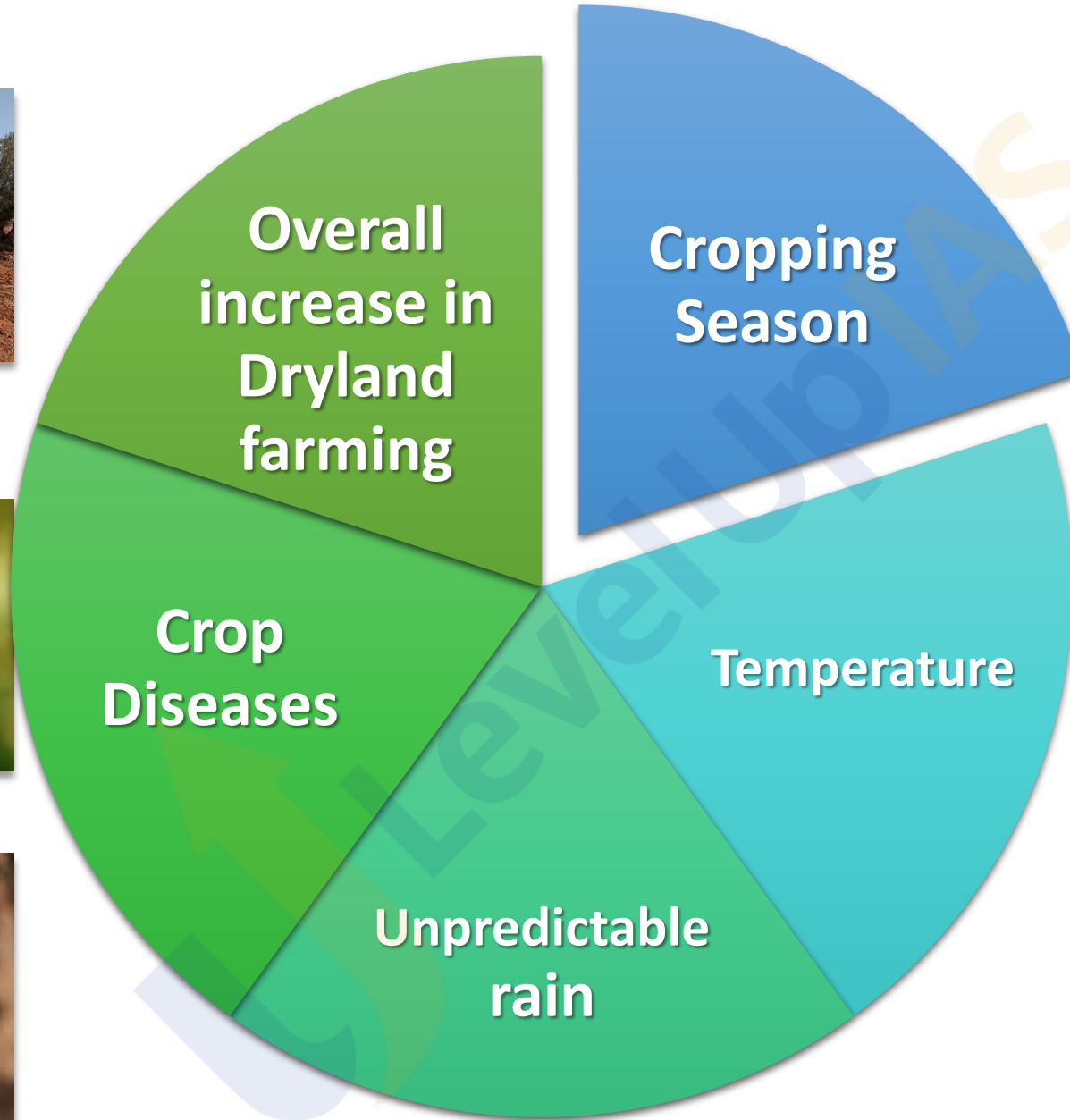
**Tropical evergreen
and semi evergreen
forests**

**Littoral and Swamp
forest**



Monsoon Variability





Questions

2015: What characteristics can be assigned to **monsoon climate** that succeeds in feeding more than 50 percent of the world population residing in Monsoon Asia?

2015: How far do you agree that the behaviour of the **Indian monsoon** has been changing due to humanizing landscapes? Discuss.